TOXICOLOGY

IS-MSN

Dr. Mookerjee

OSHA: established (as the Bureau of Labor Standards) following the "radium girls" factory workers history of the 1920s-30s
would recommend: "The Radium Girls: The Dark Story of America's Shining Women" by Kate Moore

RECOMMENDED RESOURCES FOR READING

- Katzung: Basic and Clinical Pharmacology, 15e
  - Section IX: Toxicology
    - Chapter 56: Introduction to Toxicology: Occupational & Environmental
    - Chapter 57: Heavy Metal Intoxication & Chelators
    - Chapter 58: Management of the Poisoned Patient
    - bread and butter of topic summary and backbone of these slides; comprehensive lists of drug classes- Ch 58 covers drug tox AND envt tox
- Goldfrank's Toxicologic Emergencies, 11e
  - yes, a whole textbook on this single two-hour session- not necessary to read but great for individual toxin lookup, also some good practice questions and case studies (one excerpted here)
- FYI:
  - Rang and Dale's Pharmacology: Ch 58: Harmful Effects of Drugs
    - good discussion of teratogenicity
  - Brody's Human Pharmacology: Ch 74: Essentials of Toxicology
    - organizes topic by tissue, not compounds- helpful for systems-based comparisons
    - also discussion of plasticizers (e.g., BPA, BPS) missing from other resources

All these books can be found at: https://libguides.tu.edu/COMbooks/pharmacology

…but you don't have to take my word for it….

LECTURE OBJECTIVES

- 1. (Ch 58) Explain principles of TOXICOkinetics and TOXICOdynamics based on your knowledge of PHAMACOkinetics and PHARMACOdynamics
- 2. (Ch 58) Understand the steps and significance of initial management of the poisoned patient
- 3. (Ch 58) Know antidotes to common poisons (table 4-1 in Goldfrank's)
- 4. (Ch 56) Describe major mechanisms of toxicity in acute and chronic environmental toxin exposure
- 5. (Ch 56) Know some specific common air pollutants, herbicides, and pesticides, including major mechanisms of toxicity and treatments, if any
- 6. (Ch 56) Understand the rationale and mechanisms of action for treatment of exposure to common air pollutants, herbicides, and pesticides
- 7. (Ch 57) Recognize symptoms associated with exposure to lead, arsenic, and mercury
- 8. (Ch 57) Explain mechanisms of toxicity of lead, arsenic, and mercury
- 9. (Ch 57) Explain how chelators act to effectively treat heavy metal exposure

DRUGS TO KNOW: TOXINS/STATES and ANTIDOTES

Therapeutics[b]	Indications
***Acetylcysteine* (p. 492)**	Acetaminophen and other causes of hepatotoxicity
Activated charcoal (p. 76)	Adsorbent xenobiotics in the GI tract
Antivenom (*Centruroides* spp) (p. 1563)	Scorpion envenomation
Antivenom (*Crotalinae*) (p. 1627)	Crotaline snake envenomations
Antivenom *(Micrurus fulvius)* (p. 1631)	Coral snake envenomations
Antivenom (*Latrodectus mactans*) (p. 1559)	Black widow spider envenomations
Antivenom (*Synanceja* spp) (p. 1578)	Stonefish envenomation
Atropine (p. 1503)	Bradydysrhythmias, cholinesterase inhibitors (organic phosphorus compounds, physostigmine) muscarinic mushrooms (*Clitocybe, Inocybe*) ingestions
Benzodiazepines (p. 1135)	Seizures, agitation, stimulants, ethanol and sedative–hypnotic withdrawal, cocaine, chloroquine, organic phosphorus compounds
Botulinum antitoxin (Heptavalent) (p. 586)	Botulism
Calcium chloride, calcium gluconate (p. 1403)	Fluoride, hydrofluoric acid, ethylene glycol, CCBs, hypermagnesemia, β-adrenergic antagonists, hyperkalemia

Therapeutics[b]	Indications
L-Carnitine (p. 732)	Valproic acid: hyperammonemia
Cyanide kit (nitrites, p. 1698; sodium thiosulfate, p. 1698)	Cyanide
Cyproheptadine (p. 1001)	Serotonin toxicity
Dantrolene (p. 1029)	Malignant hyperthermia
Deferoxamine (p. 676)	Iron, aluminum
Dextrose in water (50% adults; 20% pediatrics; 10% neonates) (p. 707)	Hypoglycemia
Digoxin-specific antibody fragments (p. 977)	Cardioactive steroids
Dimercaprol (British anti-Lewisite [BAL]) (p. 1251)	Arsenic, mercury, gold, lead
Diphenhydramine (p. 741)	Dystonic reactions, allergic reactions
DTPA (p. 1779) calcium trisodium pentetate	Radioactive isotopes; americium, curium, plutonium
Edetate calcium disodium (calcium disodium versenate, $CaNa_2EDTA$) (p. 1315)	Lead, other selected metals
Ethanol (p. 1440)	Ethylene glycol, methanol, diethylene glycol
Flumazenil (p. 1094)	Benzodiazepines
Folinic acid (p. 775)	Methotrexate, methanol

Table 4-1; Goldfrank's

DRUGS TO KNOW: TOXINS/STATES and ANTIDOTES

Therapeutics[b]	Indications
Fomepizole (p. 1435)	Ethylene glycol, methanol, diethylene glycol
Glucagon (p. 941)	β-Adrenergic antagonists, CCBs
Glucarpidase (p. 782)	Methotrexate
Hydroxocobalamin (p. 1694)	Cyanide
Idarucizumab (p. 911)	Dabigatran
Insulin (p. 953)	β-Adrenergic antagonists, CCBs, hyperglycemia
Iodide (SSKI) (p. 1775)	Radioactive iodine (I^{131})
Lipid emulsion (p. 1004)	Local anesthetics
Magnesium sulfate injection (p. 876)	Cardioactive steroids, hydrofluoric acid, hypomagnesemia, ethanol withdrawal, torsade de pointes
Methylene blue (1% solution) (p. 1713)	Methemoglobinemia, ifosfamide, vasoplegic syndrome, shock
Naloxone (p. 538)	Opioids, clonidine
Norepinephrine (p. 950)	Hypotension
Octreotide (p. 713)	Insulin secretagogue induced hypoglycemia
Oxygen (Hyperbaric) (p. 1676)	Carbon monoxide, cyanide, hydrogen sulfide
D-Penicillamine (p. 1215)	Copper

Therapeutics[b]	Indications
Phentolamine (p. 1129)	Vasoconstriction: cocaine, MAOI interactions, epinephrine, and ergot alkaloids
Physostigmine (p. 755)	Anticholinergics
Polyethylene glycol electrolyte lavage solution (p. 83)	Decontamination
Pralidoxime (p. 1508)	Acetylcholinesterase inhibitors (organic phosphorus compounds and carbamates)
Protamine (p. 919)	Heparin anticoagulation
Prussian blue (p. 1357)	Thallium, cesium
Pyridoxine (vitamin B_6) (p. 862)	Isoniazid, ethylene glycol, gyromitrin-containing mushrooms
Sodium bicarbonate (p. 567)	Ethylene glycol, methanol, salicylates, cyclic antidepressants, methotrexate, phenobarbital, quinidine, chlorpropamide, class I antidysrhythmics, chlorophenoxy herbicides, sodium channel blockers – know what the desired outcome of bicarb infusion is and why/how it applies to these agents
Starch (p. 1371)	Iodine
Succimer (p. 1309)	Lead, mercury, arsenic
Thiamine (vitamin B_1) (p. 1309)	Thiamine deficiency, ethylene glycol, chronic ethanol consumption ("alcoholism")
Uridine triacetate (p. 789)	Fluorouracil, capecitabine
Vitamin K_1 (p. 915)	Warfarin or rodenticide anticoagulants

Table 4-1; Goldfrank's

LECTURE OUTLINE

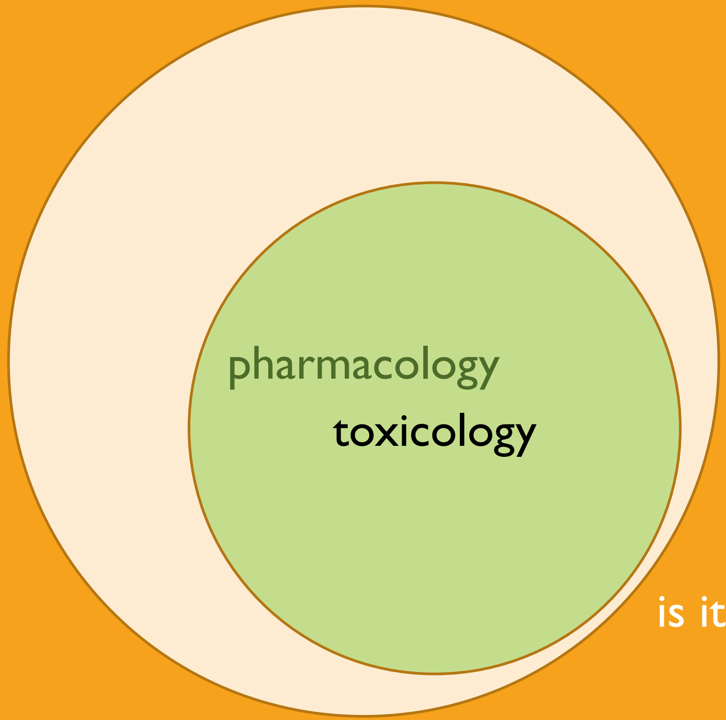
Drug lists (slides 4-5: on your own)

Drug toxicity: managing the poisoned patient (Katzung 58, Goldfrank's)

Individual common toxins (on your own)

Occupational Tox: Common air and envt pollutants (Katzung 56)

Heavy metals (Katzung 57)



is it a little cramped in here?

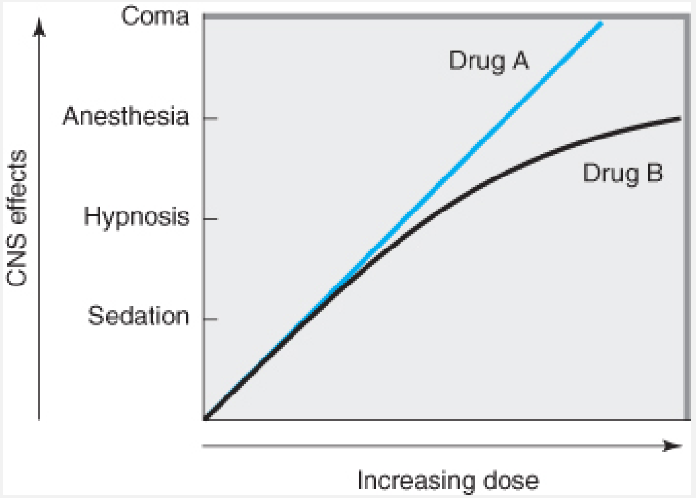
TOXICOKINETICS & TOXICODYNAMICS

- Toxicokinetics: the absorption, distribution, metabolism, and excretion (ADME) of toxins
  - similar to pharmacokinetics, but our focus here is clearing the toxin as rapidly as possible, not engineering dosage and frequency to maintain peak plasma concentration

study note: recall pharmacokinetics: same principles, different application here

- Toxicodynamics: *the effects of toxins on vital function*
  - Dose- response (therapeutic effects vs. toxic effects)
    - for drugs whose toxic effects are on-target
  - Metabolite or off-target effect
    - drugs or metabolite products with off-target toxic effects

Q: which drug is more toxic?



see next slide for A

SPECIAL CONSIDERATIONS OF TK

- **absorption**
  - peak plasma concentration could be delayed due to drug formulation, gastric emptying, etc
  - epithelial tissue injury by the toxin could speed up absorption
  - free toxin concentation past the saturation of any plasma protein binding

study q: how do you predict/diagnose toxic exposure based on timeline of ingestion?

- **distribution**
  - High V_D = extensive tissue distribution
  - low V_D = extensive plasma concentration

study q: would a high V_D or low V_D toxin be better treated with e.g., hemodialysis?

- **metabolism**
  - Is Type I or Type II metabolism impaired by toxin in a way that would slow detoxification?
  - if first-pass metabolism is impaired, how much more toxin enters circulation than expected?

type I - redox, type II - conjugation/modification

study q: what effect would this have on toxin $t_{1/2}$?

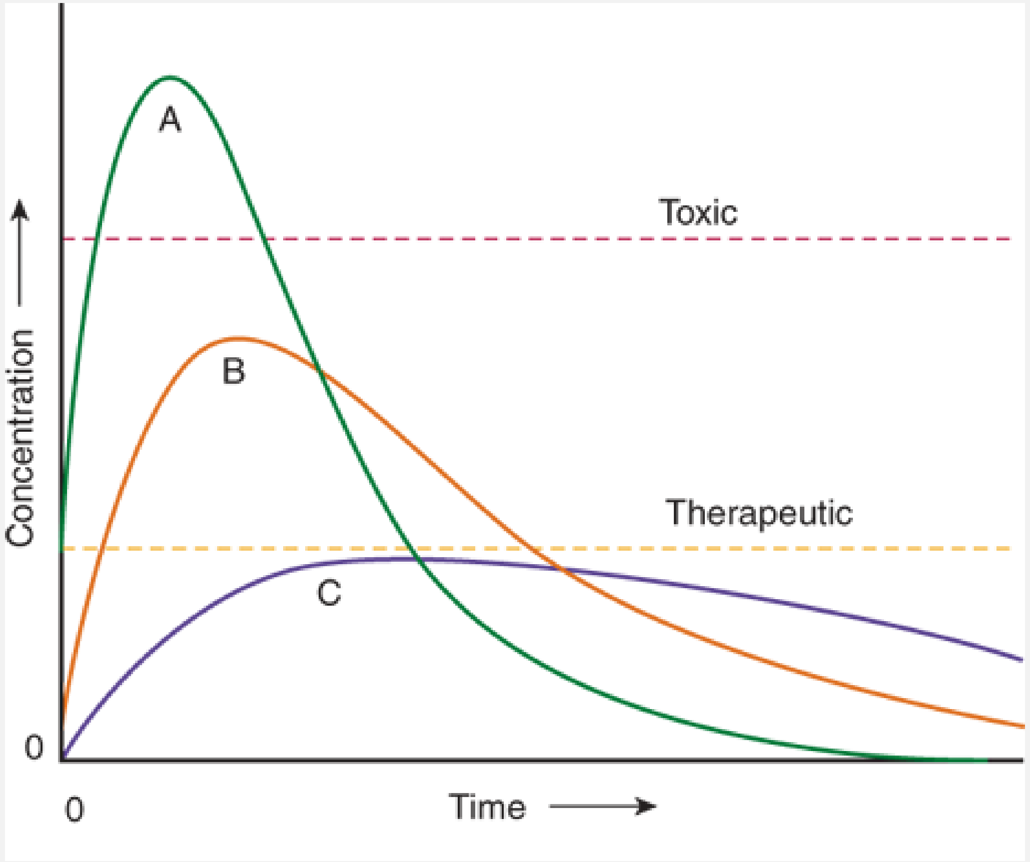
- **excretion**
  - When might zero-order elimination predominate over first-order elimination?

study note: recall zero-order vs first-order decay kinetics, and explain why zero-order decay prolongs overall toxin exposure

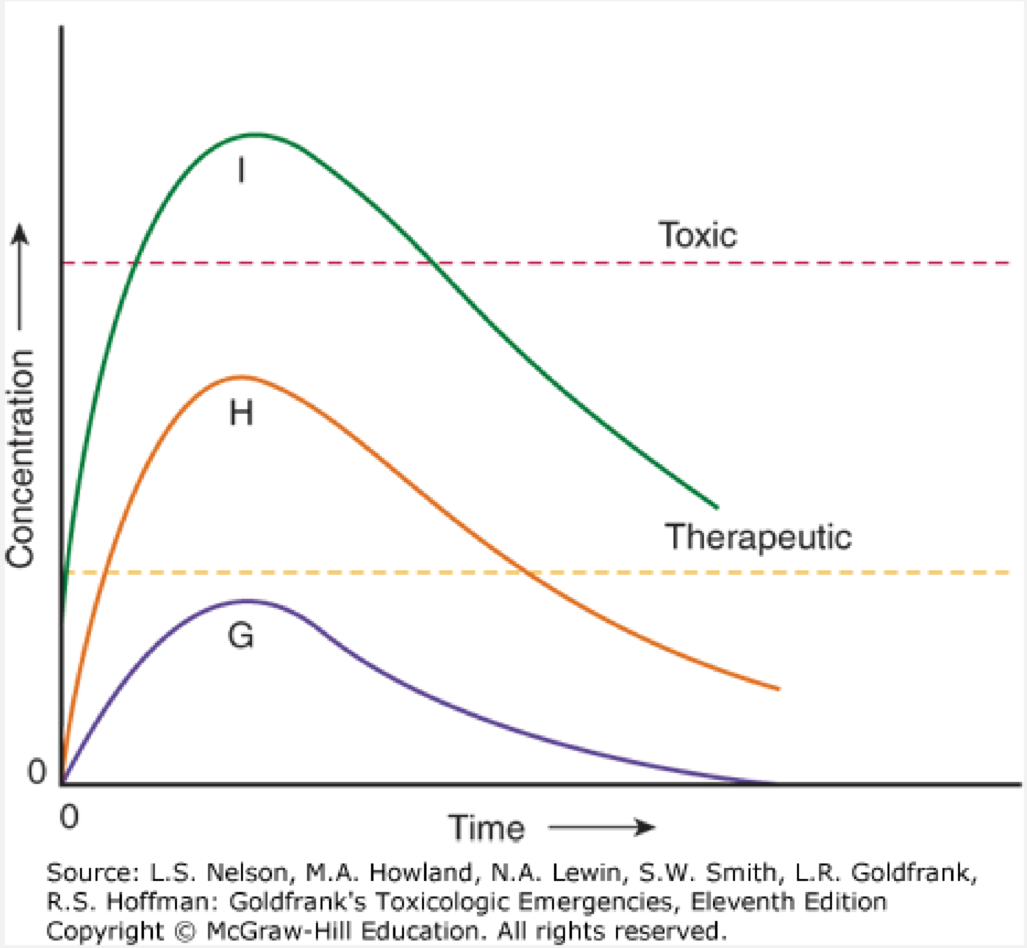
A: it depends. At low doses, they are the same. At higher doses, drug A is more toxic; i.e., you see greater symptom severity at same drug concentration

EFFECTS OF CHANGING KINETICS ON DYNAMICS

e.g., how changing the *rate* and *extent* of uptake influences therapeutic and toxic effects of medication



A, B, C: SAME bioavailability, DIFFERENT rates of absorption



G, H, I: DIFFERENT bioavailability, SAME rates of absorption

COMMON DRUGS INVOLVED IN TOXIDROMES

Bradycardia (PACED)
- Propranolol or other beta-blockers, poppies (opiates), propafenone, phenylpropanolamine
- Anticholinesterase drugs
- Clonidine, calcium-channel blockers
- Ethanol or other alcohols
- Digoxin

Tachycardia (FAST)
- Free base or other forms of cocaine
- Anticholinergics, antihistamines, amphetamines
- Sympathomimetics (cocaine, amphetamines), solvent abuse
- Theophylline

Hypothermia (COOLS)
- Carbon monoxide
- Opiates
- Oral hypoglycemics, insulin
- Liquor
- Sedative-hypnotics

Hyperthermia (NASA)
- Neuroleptic malignant syndrome, nicotine
- Antihistamines
- Salicylates, sympathomimetics
- Anticholinergics, antidepressants

Hypotension (CRASH)
- Clonidine, calcium-channel blockers
- Reserpine or other antihypertensive agents
- Antidepressants, aminophylline
- Sedative-hypnotics
- Heroin or other opiates

Hypertension (CT SCAN)
- Cocaine
- Thyroid supplements
- Sympathomimetics
- Caffeine
- Anticholinergics, amphetamines
- Nicotine

Rapid respiration (PANT)
- PCP, paraquat, pneumonitis (chemical)
- ASA and other salicylates
- Non-cardiogenic pulmonary edema
- Toxin-induced metabolic acidosis

Slow respiration (SLOW)
- Sedative-hypnotics (including GHB)
- Liquor
- Opiates, sedative-hypnotics
- Weed (marijuana)

don't memorize for now; become familiar with common toxicities resulting from medications

FYI

Table 3. Agents That Affect Pupil Size.

Miosis (COPS)
- Cholinergics, clonidine
- Opiates, organophosphates
- Phenothiazines, pilocarpine
- Sedative-hypnotics

Mydriasis (AAAS)
- Antihistamines
- Antidepressants
- Atropine and other anticholinergics
- Sympathomimetics

Table 4. Agents That Cause Seizures.

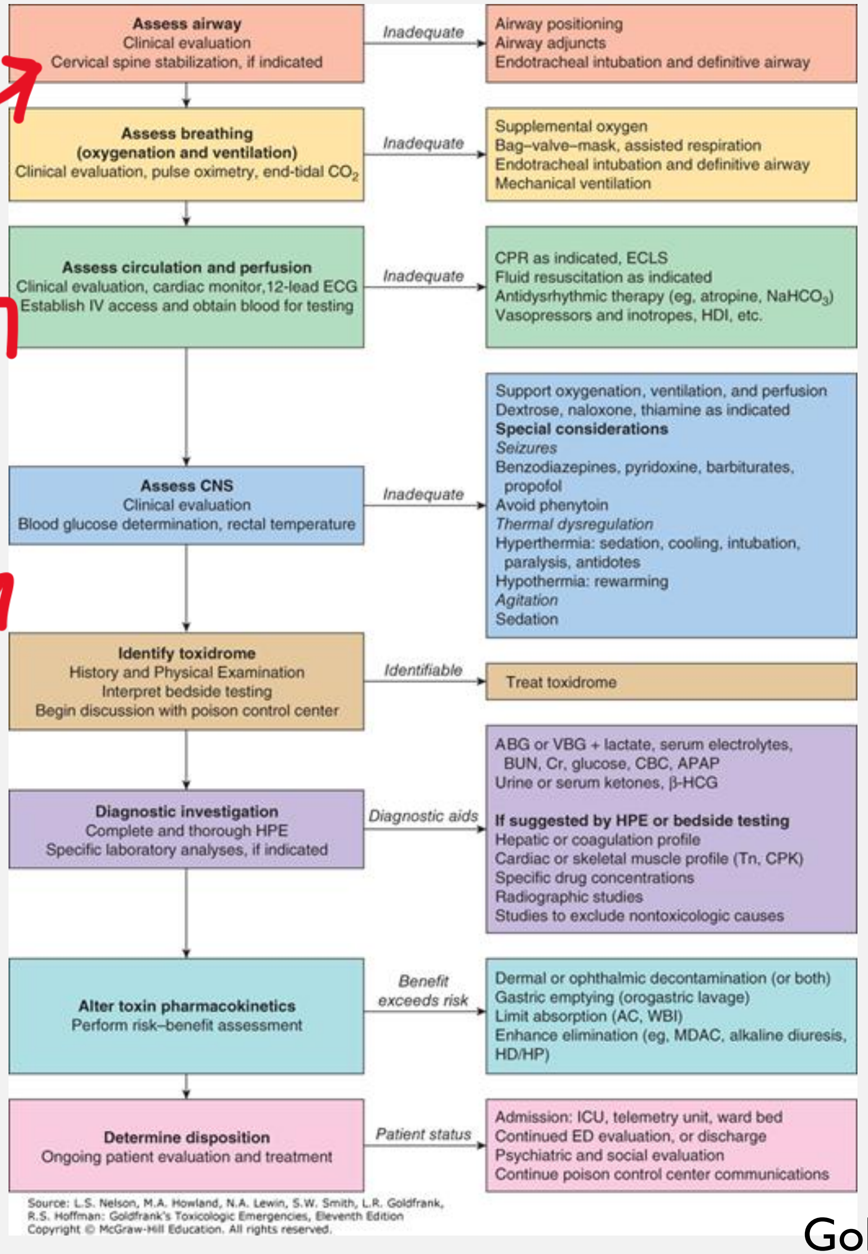
OTIS CAMPBELL*
- Organophosphates
- Tricyclic antidepressants
- Isoniazid, insulin
- Sympathomimetics
- Camphor, cocaine
- Amphetamines, anticholinergics
- Methylxanthines (theophylline, caffeine)
- Phencyclidine (PCP)
- Benzodiazepine withdrawal, botanicals (water hemlock), GHB
- Ethanol withdrawal
- Lithium, lidocaine
- Lead, lindane

FYI

http://www.ebmedicine.net/topics.php?paction=showTopicSeg&topic_id=69&seg_id=1320

APPROACH TO THE POSIONED PATIENT

- **Know the *common mechanisms* of death due to poisoning**
  - **Airway obstruction or respiratory arrest:** narcotics & sedative-hypnotic drugs
  - **Cardiovascular toxicity** (e.g., hypotension, hypovolemia, hyperthermia or hypothermia, lethal arrhythmia)**:** cardioactive drugs, opioids, antihistamines & tricyclic antidepressants
  - **Cellular hypoxia** (e.g., hypotension, lactic acidosis, tachycardia): cyanide, hydrogen sulfide, CO
  - **Seizures:** antidepressant, isoniazid, diphenhydramine, amphetamine & cocaine
  - ***Delayed organ/system damages*:** pulmonary fibrosis (paraquat), hepatic necrosis (acetaminophen)



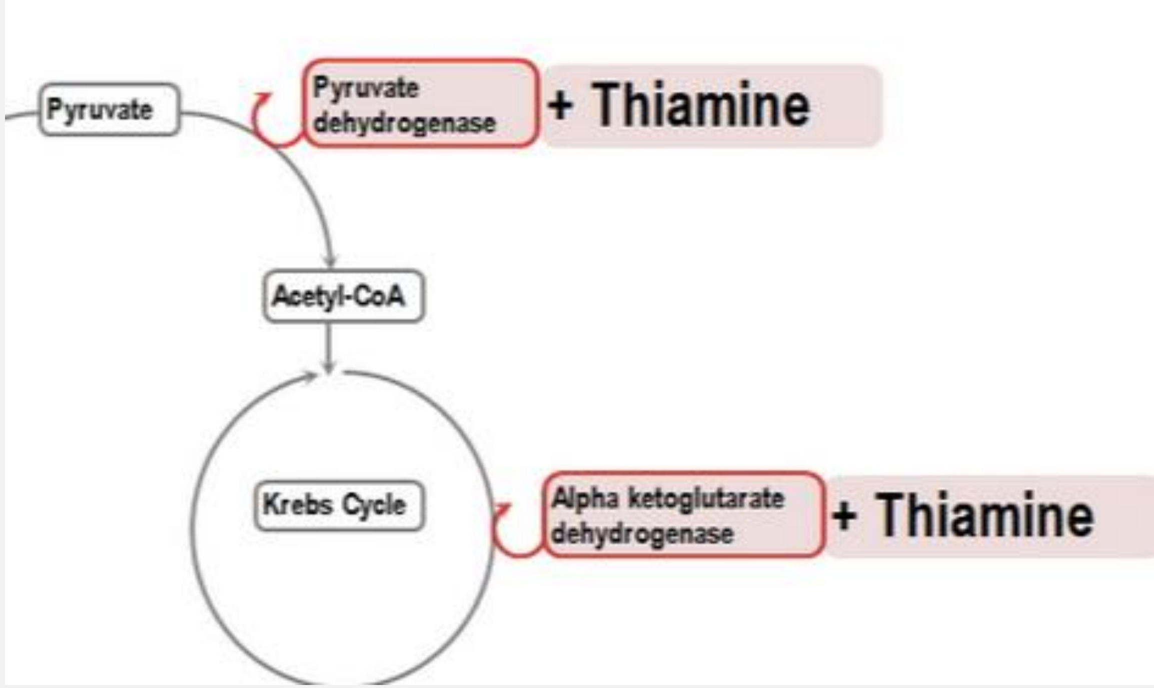
Goldfrank's

Key: symptoms: likely or possible exposure

INITIAL MANAGEMENT OF THE POSIONED PATIENT

- **Coma, seizures, or altered mental status should be immediately approached in the same way, *regardless* of the poison involved**
  - **ABCDs**
    - **Airway cleared**
    - **Breathing assessed**
    - **Circulation** (blood pressure, pulse rate, urinary output,
    - & serum glucose)
    - **Dextrose** IV (for altered mental status & hypoglycemic)
- **Thiamine** IM for alcoholic or malnourished pt
- **Naloxone** IV (*opioid* antagonist) reverse CNS
  - & respiratory depression caused by opioid drugs

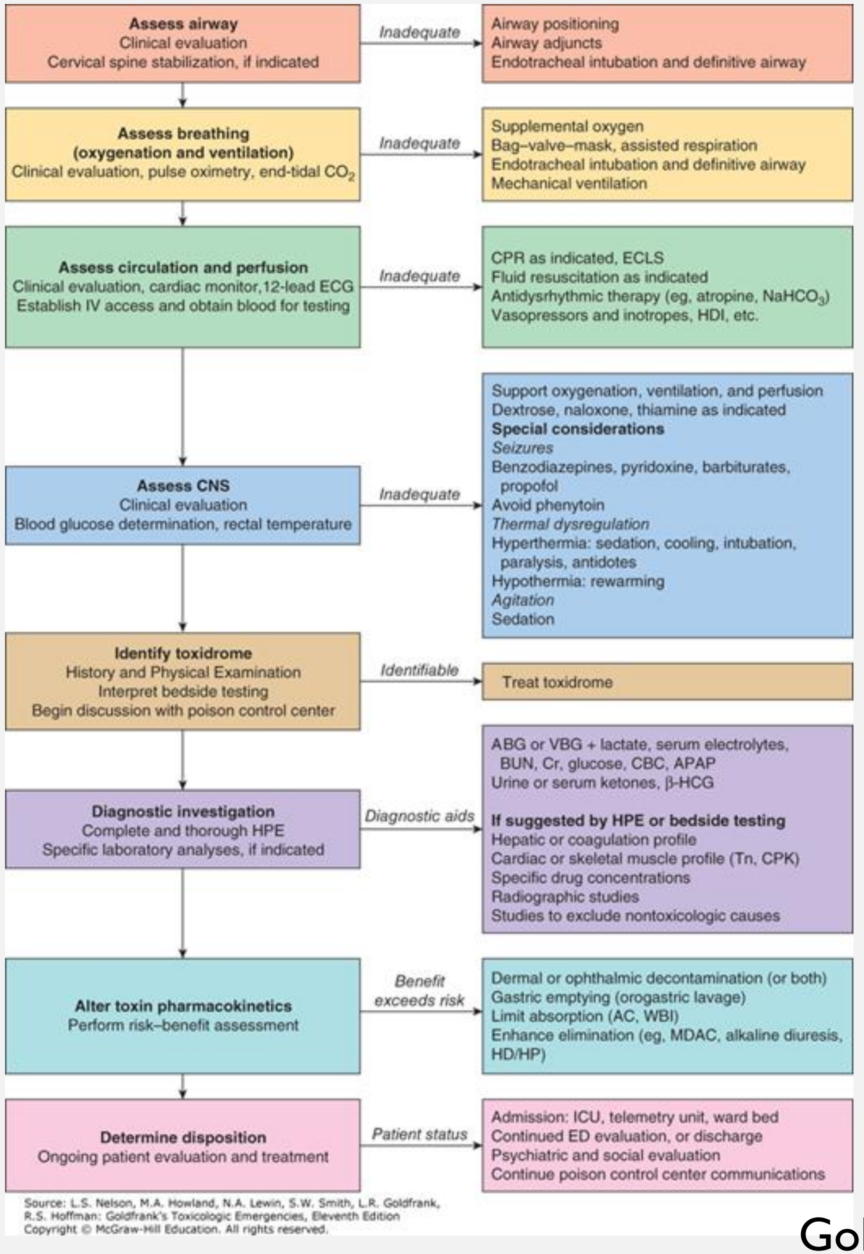
so important!



E.g., thiamine supplementation supports pyruvate dehydrogenase complex activity, allowing brain ATP production at rates required for proper function

INITIAL MANAGEMENT OF THE POSIONED PATIENT

- **History & physical examination**
  - *Other* causes of coma or seizures should be examined
  - **History-** note this can be unreliable, esp when obtained directly from poisoned pt
  - **Physical examination**
    - **Vital signs and possible toxins**
      - Hypertension & tachycardia: amphetamines, cocaine,
      - anticholinergic drug
      - Hypotension, bradycardia: calcium channel blockers,
      - β blockers, sedative hypnotics
      - Hypotension w/ tachycardia: β agonists, vasodilators,
      - tricyclic antidepressants
      - Rapid respirations (metabolic acidosis): salicylates, CO
      - Hyperthermia: sympathomimetics, anticholinergics
      - Hypothermia: CNS-depressant drugs



Key: symptoms: likely or possible exposure

Goldfrank's

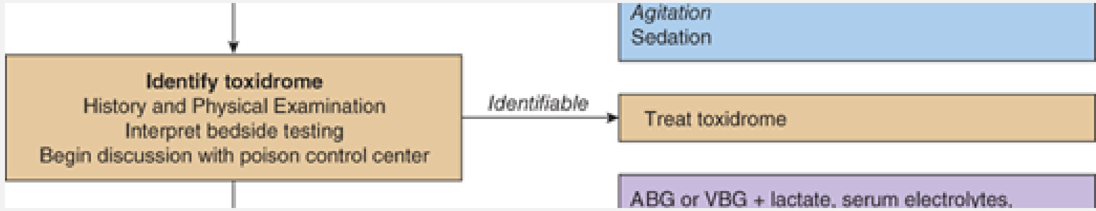
COMMON TOXIC SYNDROMES: "TOXIDROMES"

Intoxication syndrome	Substance	HR & BP	RR	Temperature	Pupils	Bowel sounds	Diaphoresis	Awareness	Additional features
Anticholinergic	• Skeletal muscle relaxants (e.g., succinylcholine, mivacurium) • Muscarinic antagonists (e.g., atropine, scopolamine, homatropine) • Certain plants of the Solanaceae family (e.g., belladonna, jimson weed)	↑	↔	↑	↑	−	−	Hypervigilance/confusion	Flushing, myoclonus, urinary retention, picking behaviour
Cholinergic	• Direct parasympathomimetics (e.g., cevimeline, oukicarpine) • Indirect parasympathomimetics (e.g., neostigmine, pyridostigmine)	↔	↔	↔	↓	+	+	Drowsiness/coma	Salivation, lacrimation, urinary incontinence, bronchoconstriction, vomiting, fasciculations
Opioid	• Full agonists (e.g., morphine, heroin, methadone) • Partial agonist (e.g., buprenorphine) • Mixed agonist/antagonists (e.g., butorphanol, balbuphine)	↓	↓	↓	↓	−	−	Drowsiness/coma	Hyporeflexia, pulmonary edema
Sympathomimetic	• Epinephrine, norepinephrine, dopamine	↑	↑	↑	↑	+	+	Hypervigilance/confusion	Tremor, hyperreflexia, seizures, bruxism
Sedative-hypnotic	• Benzodiazepines, barbiturates, gamma-hydroxybutyric acid (GHB)	↓	↓	↓	↔	−	−	Drowsiness/coma	Hyporeflexia (may mimic alcohol intoxication)
Hallucinogenic	• Marijuana, phenylcyclidine, LSD	↑	↑	↑	↑	↔	+	Hypervigilance/confusion	Hallucinations, euphoria, nausea/vomiting
Alcoholic	• Wine, beer, liquor	↑	↓	↔	↑	− / +	+	Hypervigilance/confusion → Drowsiness/coma	Reduced alertness, slurred speech, unsteady gait and posture, nausea/vomiting

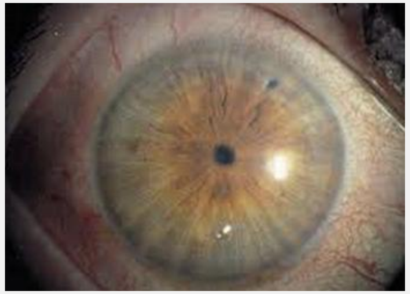
↔ Unchanged — Drowsiness/coma — Hypervigilance/confusion

AMBOSS

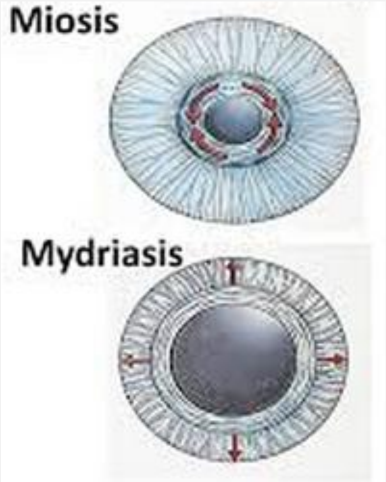
INITIAL MANAGEMENT OF THE POSIONED PATIENT



- **History & physical examination (Eye)**



Miosis:
Cholinesterase inhibitors
Heroin
Opioids
Phenothiazine



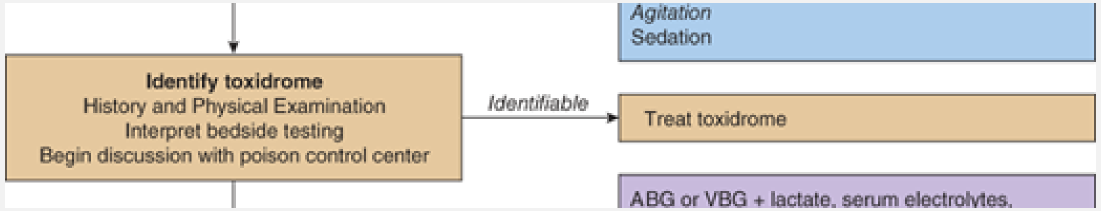
Mydriasis: stimulants
Cocaine
AmphetamineLSD
Atropine

- Horizontal nystagmus: alcohol, phenytoin, barbiturates
- Vertical & horizontal nystagmus: phencyclidine
- Ptosis and ophthalmoplegia: botulism

Key: symptoms: likely or possible exposure

INITIAL MANAGEMENT OF THE POSIONED PATIENT

- **History & physical examination (cont'd)**



- Mouth
  - Signs of burn or soot
  - Odors of hydrocarbon solvent, alc., ammonia,
  - 'almond' (cyanide poisoning)

- Skin
  - anticholinergic
  - Flushed, hot, dry: atropine, antimuscarinics
  - cholinergic crisis
  - Excessive sweating: organophosphates, nicotine
  - insufficient oxygen - cellular hypoxia
  - Cyanosis: hypoxemia (CO) or methemoglobinemia
  - Icterus: hepatic necrosis due to acetaminophen

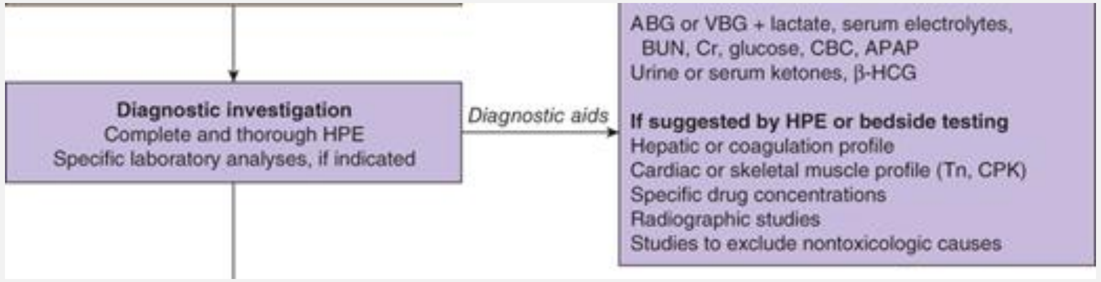
- Abdomen
  - Ileus: antimuscarinic, <u>opioid</u>, sedative drugs
  - Cramp/diarrhea: organophosphates, <u>iron</u>, theophylline

- Nervous system
  - Ataxia & dysarthria: phenytoin, carbamazepine, <u>alcohol</u>
  - Twitching: <u>cocaine, atropine, anticholinergics</u>
  - Muscular rigidity: <u>haloperidol</u> & other antipsychotics
  - Seizures: antidepressants, <u>cocaine, amphetamines,</u>
  - <u>isoniazid</u>, theophylline, diphenhydramine
  - Flaccid coma w/ absent reflexes: sedative-hypnotics

Key: symptoms: <u>likely</u> or possible exposure

LABORATORY & IMAGING PROCEDURES

- **Arterial blood gases**
  - ↓PO_2 (hypoxia): hyperventilation, drug-induced pulmonary edema
  - Poor tissue oxygenation (hypoxia, hypotension, cyanide):
    - may lead to *metabolic acidosis*
- **Electrolytes**
  - **Anion gap = $(Na^+ + K^+) - (HCO_3^- + Cl^-)$**
  - ↑ **anion gap**: **metabolic acidosis** (DK, renal failure, etc),
  - aspirin, metformin, MeOH, ethylene glycol, isoniazid, iron
  - **Hyperkalemia**: K^+, β blockers, digitalis, K^+-sparing diuretics, fluoride
  - **Hypokalemia**: barium, β agonist, caffeine, theophylline, thiazide & loop diuretics

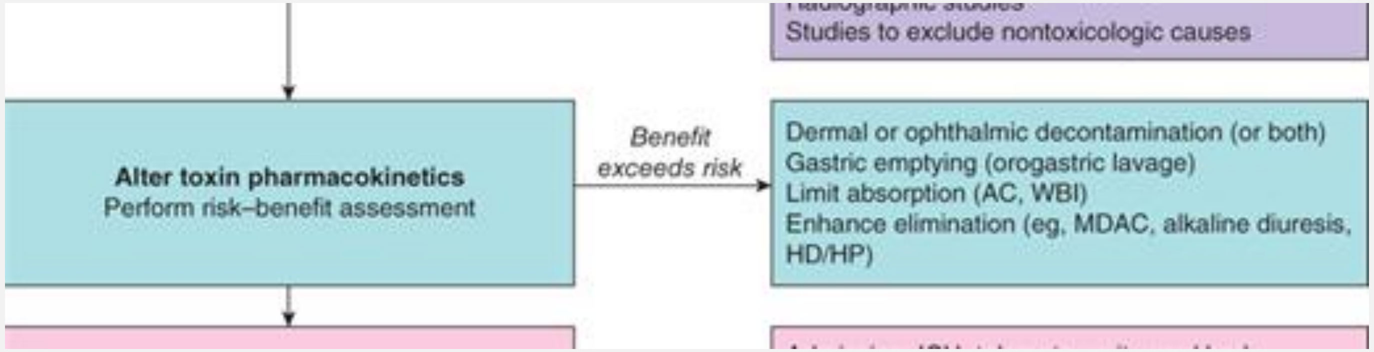


- **Renal function tests**
  - Direct or indirect effect (e.g., shock or myoglobinuria)
  - Measure blood urea nitrogen & creatinine levels
  - ↑ serum CK & myoglobin in the urine: muscle necrosis due to seizures or muscular rigidity
  - ↑ oxalate crystals in the urine: ethylene glycol poisoning
- **Electrocardiogram**
  - Widening QRS or QT interval prolongation: tricyclic antidepressant or quinidine overdose
  - AV blocks: digoxin
- **Imaging findings**
  - *Radiopaque* tablets (e.g., iron, potassium)
  - CT scan: head trauma
  - Chest radiographs: pulmonary edema, aspiration pneumonia

Key: symptoms: likely or possible exposure

TREATMENT APPROACHES

- **Decontamination**
  - **Skin**
  - **GI tract**
    - **Emesis**
    - **Gastric lavage**
    - **Activated charcoal** (adsorb or sequester drugs/toxins- 'gut dialysis')
    - **Cathartics** (laxative)



- **Specific antidotes**

- **Enhance elimination of toxins**
  - **Dialysis**
    - **Peritoneal dialysis** (simple but less effective)
    - **Hemodialysis** (more effective)
  - **Forced diuresis & urine pH manipulation**
    - Forced diuresis may cause electrolyte imbalance
    - ↑urine pH is useful for salicylate poisoning

$$HA \rightleftharpoons H^+ + A^-$$

HA can cross plasma membrane (and re-enter circ, A^- cannot

CASE EXAMPLE: ON YOUR OWN

- **History**
- A 22-year-old man was brought to the emergency department after being found unconscious at a movie theater. In the field, the paramedics inserted an intravenous (IV) line, performed a rapid reagent glucose test that was recorded as 88 mg/dL, and administered oxygen via nasal cannula at 4 L/min when his room air pulse oximetry was determined to be 91%. The patient had no medical records at the receiving hospital, and no useful information was found among his belongings.
- **Physical Examination**
- On arrival at the ED, the patient was unconscious, with no response to physical stimuli. Vital signs were: blood pressure, 92/56 mm Hg; pulse, 52 beats/min; respiratory rate, 10 breaths/min; temperature, 97.4°F (36.3°C) {rectal}; and oxygen saturation, 98% on 4 L O_2/min. A repeat rapid reagent glucose was unchanged. Examination was notable for 2- to 3-mm pupils that responded to light, normal oculocephalic testing, nearly flaccid muscle tone, and downgoing toes bilaterally. His head was without signs of trauma, and his neck was supple. Examination of the chest revealed scattered coarse breath sounds and a regular heart rhythm with normal heart sounds and no murmurs. The abdomen was soft, bowel sounds were present, and no abnormalities were noted on the skin or extremities. When oxygen was removed, his saturation fell to 92%, and a continuous end-tidal CO_2 monitor measured 48 mm Hg.

reference ranges:
pulse/ox: 95% or higher
blood glucose: fasting, 70-99
vitals:
- BP: 90/60 – 120/80
- pulse: 60-100/min
- resp: 12-16/min
- pupils: 2-8 mm
- pCO2: 35-45mmHg

From: Goldfrank's, Case 2

QUICK REVIEW: CASE EXAMPLE

- Given the patient's initial presentation, what first response is warranted?

  insufficient oxygen - give O2

- What should happen next?

  improve o2 sat

- What tests might be recommended after the patient it stabilized?

  ABG, electrolytes

QUICK REVIEW: CASE EXAMPLE ANSWER SUMMARY

- Given the patient's initial presentation, what was the first response?
  - Given the patient's hypoventilation and small pupils, graded doses of naloxone were given
    - why? small pupil size, esp in bright light, consistent with opioid toxicity. naloxone can also treat clonidine toxicity (indirect mechanism)
    - outcome: no response

- What should happen next?
  - ABCs: is airway obstructed? no. Is pt breathing? not enough (pulse ox low, pCO2 high) – pt was intubated and connected to a ventilator (note that D: dextrose was not given, in part because blood glucose was in the normal range and therefore not indicative of hypoglycemic coma)

- What additional information gathering or treatment should follow?
  - includes, e.g., blood values: arterial blood gas, CBC, electrolytes, and ethanol and acetaminophen (APAP) concentrations.
  - Electrocardiography (ECG) – here, showing sinus bradycardia with normal axis and intervals, and normal ST segments and T-waves.
  - To address low BP, 1 L of 0.9% sodium chloride was infused, wih resulting BP 102/60 mm Hg, with no change in pulse.
  - To address the potential ingestion of toxin, a nasogastric tube was inserted through which 60 g of activated charcoal was instilled into the stomach.

EXTENDED CASE DISCUSSION AND EXPLANATION

Immediate Management Given the patient's hypoventilation and small pupils, graded doses of naloxone were given (0.04, 0.1, and 2 mg IV) without response (Antidotes in Depth: A4). On further examination, the patient's gag reflex was absent, prompting endotracheal intubation, which was performed without medications, and the patient was attached to a mechanical ventilator. A postintubation arterial blood gas, a complete blood count, electrolytes, and ethanol and acetaminophen (APAP) concentrations were obtained. Electrocardiography (ECG) showed sinus bradycardia with normal axis and intervals, and normal ST segments and T-waves were observed. A total of 1 L of 0.9% sodium chloride was infused, and his blood pressure increased to 102/60 mm Hg, with no change in his pulse. A nasogastric tube was inserted through which 60 g of activated charcoal was instilled into the stomach.

What Is the Differential Diagnosis?This patient was comatose, with remarkable vital signs (hypotension, bradycardia, and hypoventilation) and remarkable physical findings (miosis, coma, flaccid muscles). The differential diagnosis is extensive and includes many xenobiotics from diverse chemical classes. The most common causes are listed in Table CS2–1. In many cases, it is not necessary to establish the correct diagnosis but rather to exclude diagnoses that require specialized care or are amenable to specific interventions.

- **What Clinical and Laboratory Analyses Help Exclude Life-Threatening Causes of This Patient's Presentation?** Either a rapid reagent glucose determination should be obtained or hypertonic dextrose should be administered in every comatose patient (Chaps. 3 and 4 and Antidotes in Depth: A1 and A8). A normal blood glucose concentration or failure to respond to an appropriate dose of hypertonic dextrose essentially excludes persistent hypoglycemia. When hypoventilation is present, a graded trial of naloxone is indicated recognizing that patients who have overdosed on clonidine and similarly acting (i.e., centrally acting alpha2 agonists) antihypertensives may respond (Antidotes in Depth: A4). A screening ECG is essential and may guide diagnosis and management (Chap. 15). Prolongation of the QRS complex occurs with some antidepressants, antipsychotics, and antiepileptics, and it may indicate the need to administer hypertonic sodium bicarbonate (Antidotes in Depth: A5). In addition, prolongation of the QT interval is commonly recognized with many antipsychotic and antidepressant overdoses (Chaps. 67 and 68). Although metabolic acidosis with an elevated anion gap is not typical of the xenobiotics listed, it might be indicative of a serious coingestant. Similarly, coma is not expected after a typical APAP overdose, but APAP is commonly ingested and coformulated with sedatives and opioids.
- When a patient presents with coma, mildly abnormal vital signs, and flaccid muscles and if ECG results and glucose, oxygenation, anion gap, and APAP are normal or can be corrected, then the patient will likely do well with supportive care.

Further Diagnosis and Treatment The arterial blood gas and electrolytes showed no evidence of acidosis and a normal anion gap and a normal osmol gap. Other than bradycardia, the ECG was normal. The serum APAP concentration was 162 mcg/mL. Because the time of ingestion was unknown, *N*-acetylcysteine (NAC) was initiated (Antidotes in Depth: A3). After some time, a family member was contacted who disclosed that the patient had several previous suicide attempts and was known to be taking phenobarbital. A serum phenobarbital concentration was obtained on original blood in the laboratory and was reported to be 132 mcg/mL. A bicarbonate infusion was started to alkalinize the urine of the patient (Antidotes in Depth: A5). A 21-hour NAC regimen was completed at which time the patient's APAP was not detectable, and the aminotransferases were within normal limits. Over the next 48 hours, the patient regained consciousness and was extubated. A consultation with a psychiatrist determined the need for involuntary hospitalization, and the patient was transferred to the psychiatric service.

From: Goldfrank's, Case 2

LECTURE OUTLINE

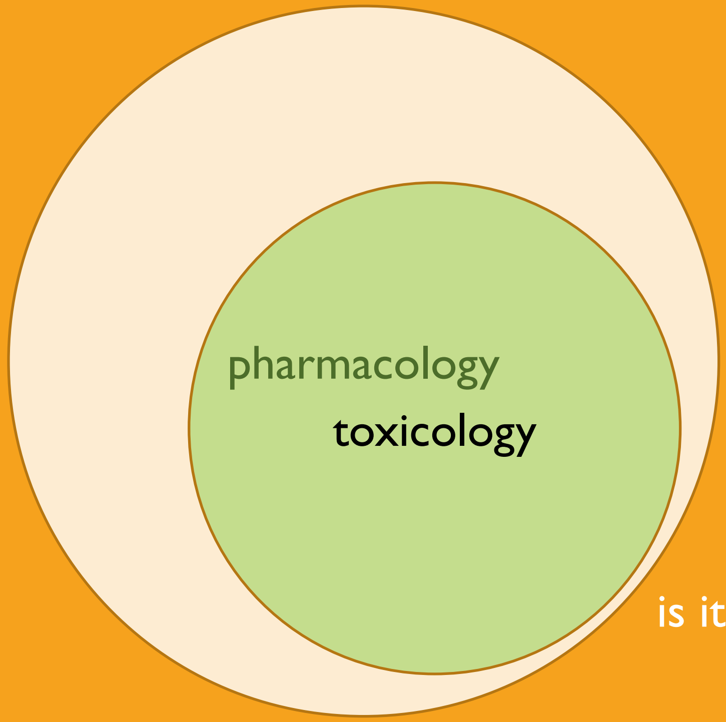
Drug lists (slides 4-5: on your own)

Drug toxicity: managing the poisoned patient (Katzung 58, Goldfrank's)

Individual common toxins (on your own): special focus on aspirin and APAP

Occupational Tox: Common air and envt pollutants (Katzung 56)

Heavy metals (Katzung 57)



is it a little cramped in here?

COMMON TOXINS: KATZUNG

- Acetaminophen
- Amphetamines & other stimulants
- Anticholinergic agents
- Antidepressants
- Antipsychotics
- Aspirin
- beta blockers
- Calcium channel blockers
- Carbon monoxide & other toxic gases
- Cholinesterase inhibitors
- Digoxin
- Ethanol & sedative-hypnotic drugs
- Ethylene glycol & methanol
- Iron & other metals
- Opioids
- snake, spider, jellyfish evenomation
- Theophylline

mostly supportive and waiting for body to clear

overall goal of this section: match toxins with treatments, be able to explain MOA of both toxicity and treatment

COMMON OVERDOSE: APAP

- **Acetaminophen**

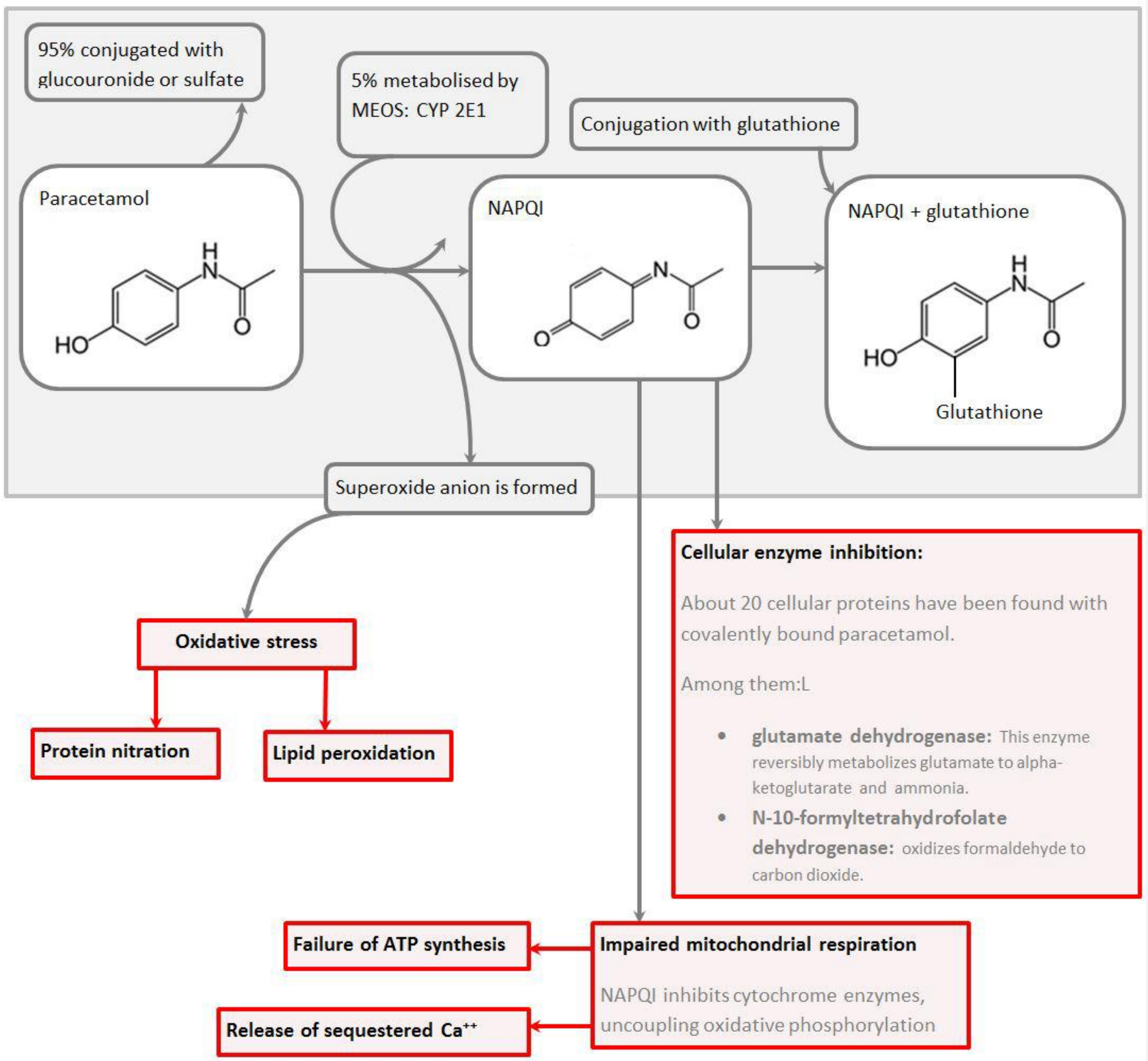
vocab corner: APAP
N-**a**cetyl-**p**-**a**mino**p**henol

- Common in suicide attempts & accidental poisonings
- > 150-200 mg/kg (children) or 7 gm total (adults)
- *Toxic metabolites* produced in the *liver* via p450 enzymes
- **Liver injury** appears within 24-36 hrs
  - ↑*aminotransferase* levels
  - *hypoprothrombinemia*
  - Severe cases: ***fulminant liver failure,* leading to *hepatic***
  - ***encephalopathy & death***
- History of alcohol abuse or concurrent use of drugs that increase CYP activity are at elevated risk of APAP toxicity

Active metabolite NAPQI : N-acetyl-p-aminobenzoquinone imine

COMMON OVERDOSE: APAP CONT'D

- with normal dose, most (95%) APAP is "safely" metabolized by type II rxns
  - glucuronidation
  - sulfation
- about 4% is converted by type I rxn (oxidation, mostly by CYPs) into NAPQI
  - GSH can handle these amounts
- with overdose:
  - glucuronidase becomes saturated; about 15% of APAP → NAPQI
  - GSH becomes saturated: NAPQI accumulates
  - both NAPQI and $O2^{\bullet -}$ disrupt cellular functions

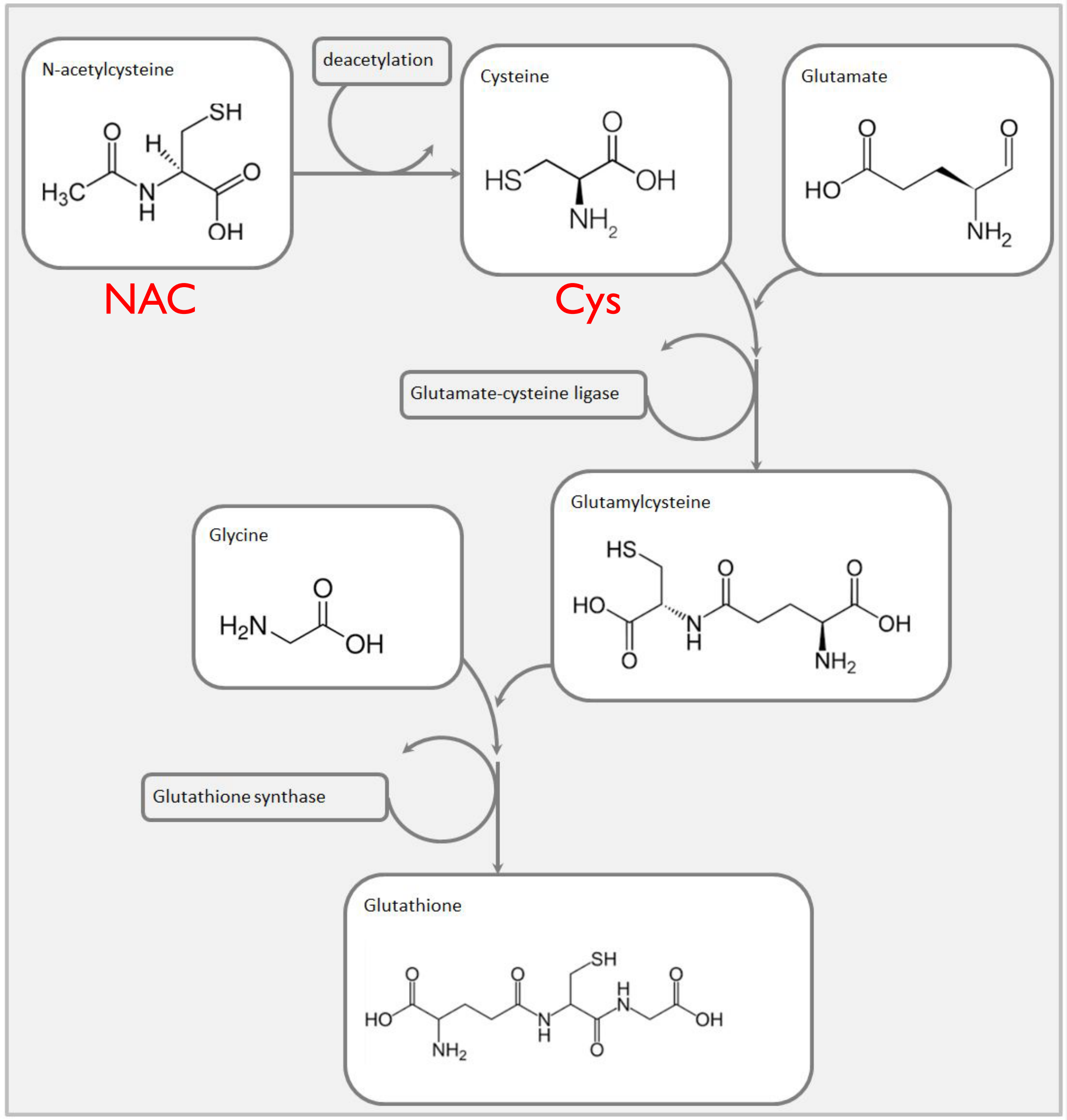


https://dmd.aspetjournals.org/content/31/12/1499.short

APAP OD ANTIDOTE: NAC

- why is GSH "depleted"?
  - oxidation to GSSG?
  - conjugation to NAPQI "uses up" available GSH?

- perhaps, but also:
  - NAPQI-GSH is exported from hepatocytes and excreted
  - this is WHY resynthesis is so important



https://dmd.aspetjournals.org/content/31/12/1499.short

FYI q to ponder: Why use NAC and not just cysteine itself? ? In theory, it would be fine. In practice, however:

- Cys oxidizes in solution and precipitates out
- Cys in soln smells and tastes bad
- therefore, NAC is easier to deliver and easier to tolerate than Cys

COMMON OVERDOSE CONT'D

- **Amphetamines & other stimulants**

- Methamphetamine
- Methylenedioxymethamphetamine (MDMA)
- Cocaine, pseudoephedrine (Sudafed), ephedrine
- Synthetic cathinones (MDPV, 'bath salts')

- As stimulants: euphoria, wakefulness, sense of power & well-being
- Higher doses: restlessness, acute psychosis, tachycardia, dehydration & eventually hypotension (due to prolonged muscular hyperactivity), seizures, hyperthermia, and rhabdomyolysis

- Supportive treatment (*no* specific antidote)
  - Ammonium chloride acidification facilitates elimination
  - Seizures: benzodiazepines (e.g., lorazepam IV)
  - Hyperthermia: neuromuscular paralysis

- **Antidepressants**

- Tricyclic antidepressants (TCAs)
  - Amitriptyline, desipramine, doxepin (lethal: >1 g)
  - Competitive antagonists at muscarinic R
  - Anticholinergic effects; vasodilation, seizure, cardiac toxicity (wide QRS, arrhythmias)
  - Supportive treatment
    - Norepinephrine IV: for hypotension
    - Sodium bicarbonate: for cardiotoxicity (Na^+ influx helps overcome Na channel block)

- Monoamine oxidase inhibitor (MAOI)
  - Tranylcypromine, phenelzine
  - Severe hypertension, drug interactions, seizure

- Newer antidepressants (e.g., SSRIs)
  - Fluoxetine, paroxetine, citalopram, venlafaxine
  - Seizures, QT prolongation, arrhythmia,
  - drug interactions (causing *serotonin syndrome* w/ MAOI)

COMMON OVERDOSE CONT'D

- **Anticholinergic agents**
  - e.g., diphenhydramine, doxylamine
- Syndrome: *sinus tachycardia, dilated pupils, agitated*
- *delirium, seizures* (for antihistamines or tricyclic antidepressants), *urinary retention*
- Supportive treatment
  - Antipsychotic agents (e.g., haloperidol): for agitation
- Specific antidote: <u>physostigmine</u> IV
  - cholinesterase inhibitor
  - Potential bradycardia and seizures
  - Avoided in pt with suspected tricyclic antidepressant OD (may exacerbate cardiotoxicity)

- **Antipsychotics**
- Phenothiazines, butyrophenones, etc.
- Syndrome: CNS depression, seizures, hypotension, QT prolongation
- Supportive treatment; activated charcoal

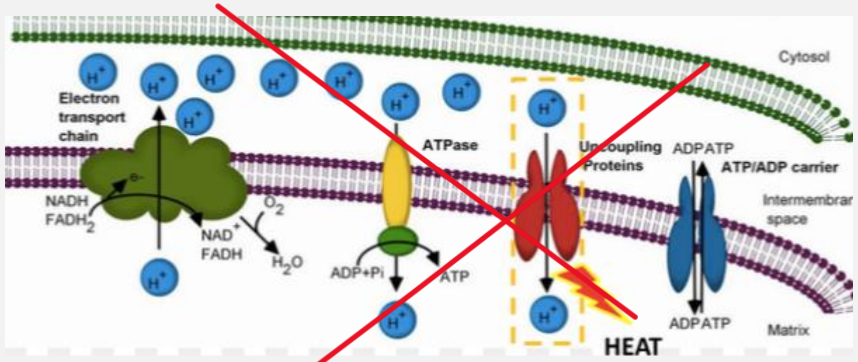
<u>Anticholinergic toxidrome</u>:
Red as a beet
hot as a hare
dry as a bone,
blind as a bat
mad as a hatter

ASPIRIN TOXICITY CONT'D

- **Aspirin**

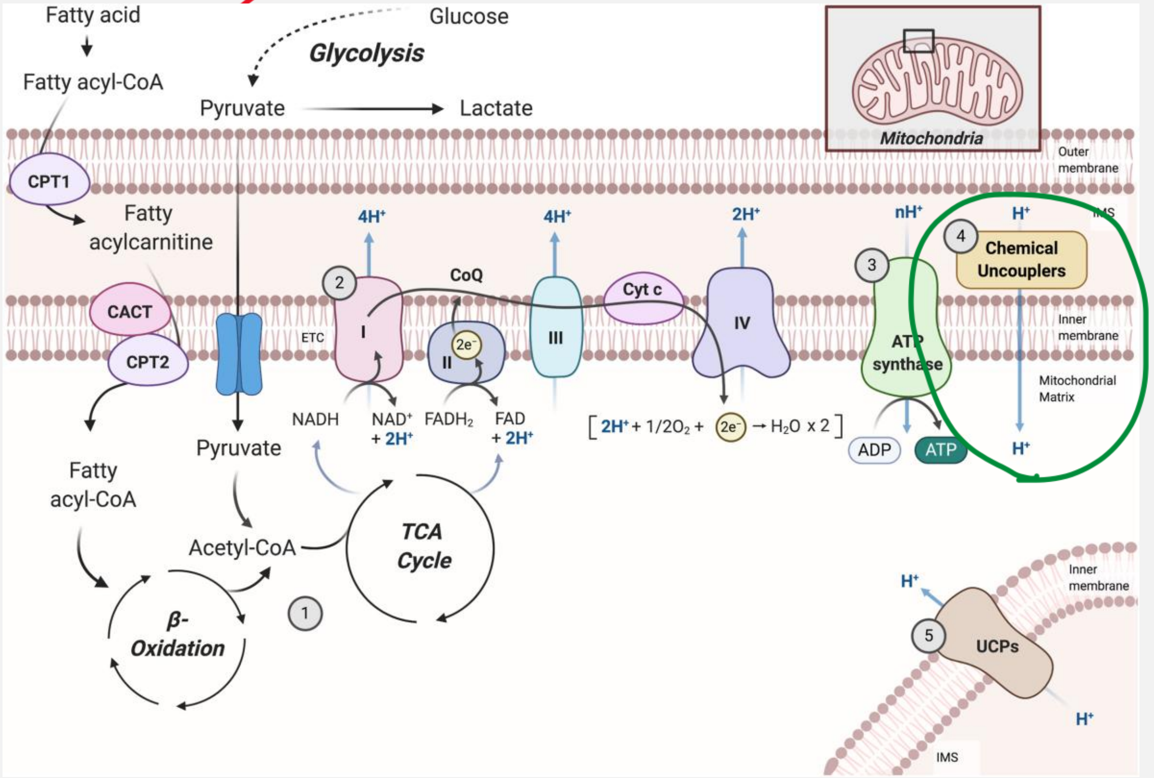
- > 200 mg/kg (acute intoxication); chronic overmedication
- Uncouples oxidative phosphorylation
- Symptoms: metabolic acidosis, hyperventilation,
  - respiratory alkalosis (uncommon in children),
- ↑anion gap, hyperthermia, vomiting, & dehydration
  - Severe: seizures, coma, pulmonary edema & cardiovascular collapse
- Care: Supportive treatment, activated charcoal, gastric lavage,
- IV sod. bicarbonate (alkaline diuresis) or hemodialysis

ASPIRIN TOXICITY: CHEMICAL UNCOUPLER

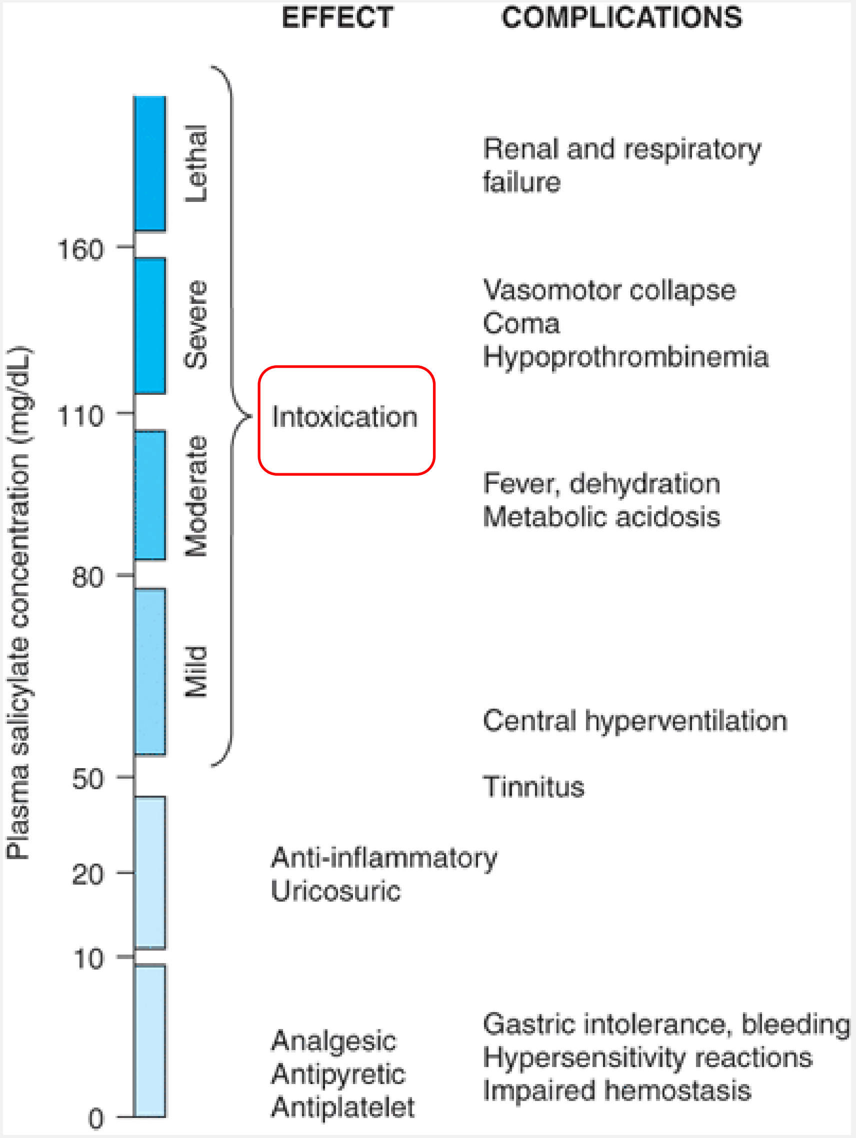


Aspirin overdose= uncoupling protein (UCP) (➔Reduced ATP, hyperthermia)

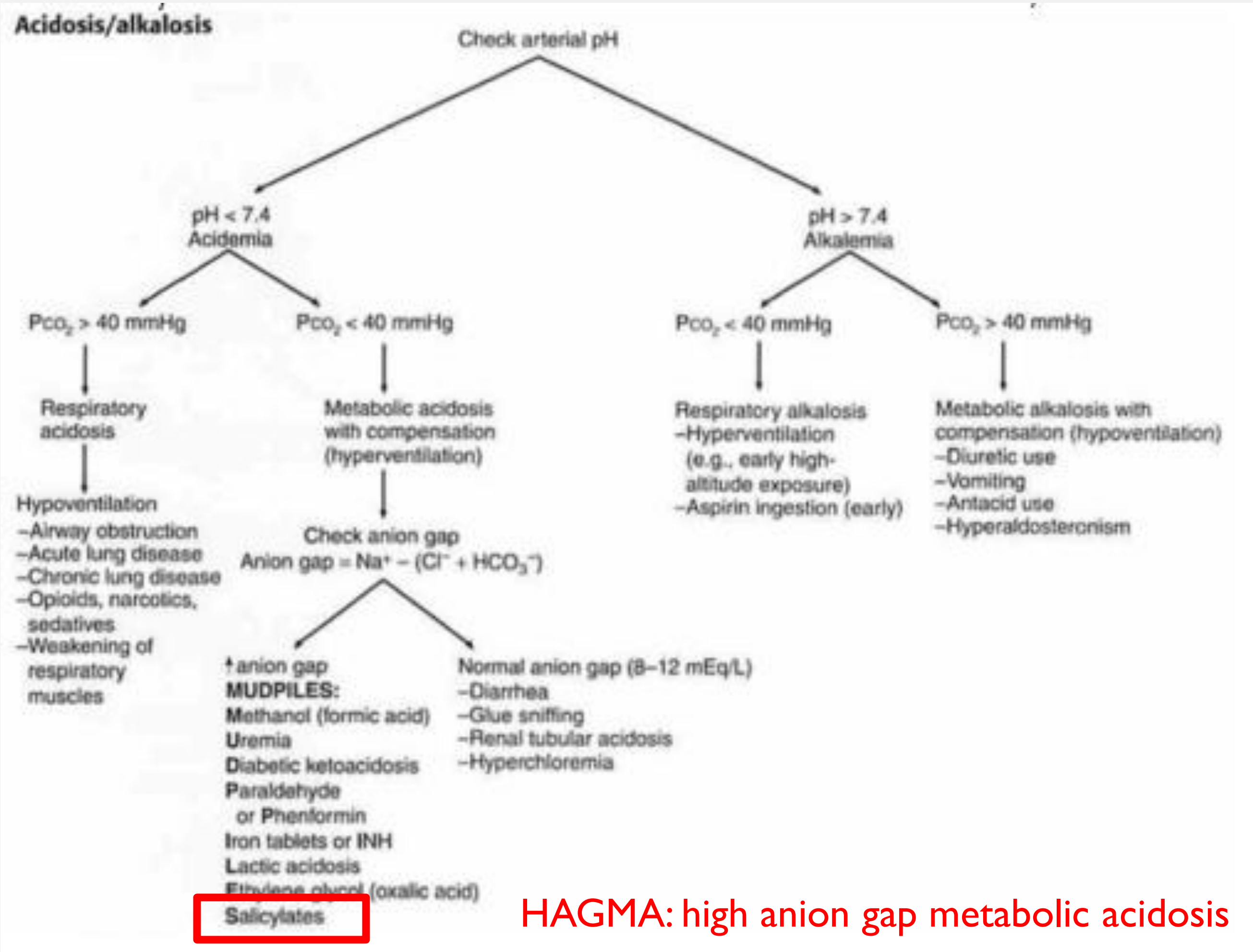
???? UCP activator? no



chemical uncoupler? YES



PH DERANGEMENT INTERPRETATION/CAUSES



HAGMA: high anion gap metabolic acidosis

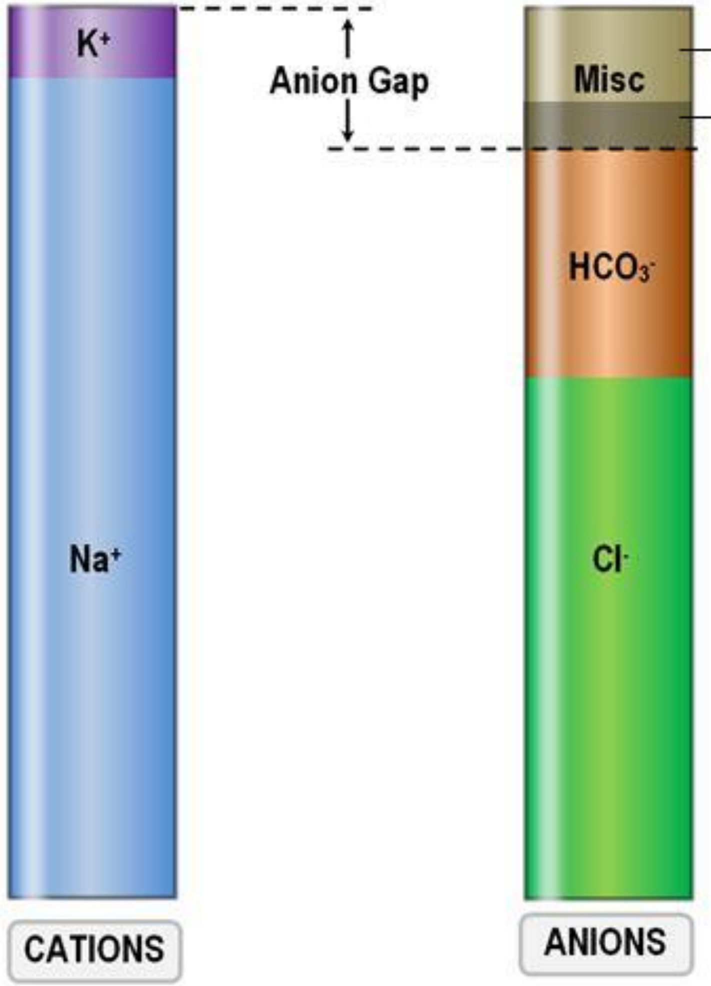
Metabolic Acidosis Diabetic ketoacidosis Diarrhea Renal failure Shock Aspirin overdose ← Sepsis	**Metabolic Alkalosis** Loss of gastric secretions Overuse of antacids K+ wasting diuretics
Respiratory Acidosis Hypoventilation COPD Airway obstruction Drug overdose Chest trauma Pulmonary edema Neuromuscular disease	**Respiratory Alkalosis** Hyperventilation Hypoxia Anxiety High altitude Pregnancy Fever

metabolic acidosis: blood acidemia with HCO_3^- <24 mEq/L

anions accumulating as salicylate and lactate

why??? salicylate itself is an anion, and mt uncoupling causes glycolysis to supply ATP, creating lactate$^-$ + H^+

FYI FOR THIS EXAM: WHAT IS THE ANION GAP?????



Contribution from the anionic charge of albumin

Miscellaneous anions, which could be...

- *L- or D-Lactate*
- *Ketones*
- *Salicylate*
- *Pyroglutamate*
- *Metabolic products of toxic alcohols:*
  - *Formate (methanol)*
  - *2-hydroxyethoxyacetate (diethylene glycol)*
  - *Oxalate (ethylene glycol)*
  - *Acetate (ethanol)*
- *Retained non-volatile acids of renal failure:*
  - *Sulfate*
  - *Phosphate*
  - *Urate*
  - *Hippurate*

large anion gap - misc is bigger than it should be (salicylate, lactate, etc.

it is a rule of thumb: quickly informative but imperfect most of the time
Assumption of electrochemical neutrality: [anions] = [cations]

AG = (Na + K) – (HCO_3 + Cl)
The "anion gap" is all the anions assumed by cation measurement but not directly measured – this is a normal feature of electrolyte and charge analysis

source and excellent review: https://journals.lww.com/CJASN/pages/articleviewer.aspx?year=2007&issue=01000&article=00026&type=Fulltext

FYI

OTHER FACTORS CAUSING INCREASED ANION GAP

↑Anion Gap	Agents
Organic acid metabolites	Methanol, ethylene glycol, diethylene glycol
Lactic acidosis	cyanide, CO, ibuprofen, isoniazid, metformin, salicylates, valproic acid; any drug-induced seizures, hypoxia, or hypotension

FYI

COMMON OVERDOSE CONT'D

- β blockers

- Propranolol is most toxic
- Symptoms:
  - Bradycardia and hypotension are most common
  - Seizures and cardiac conduction block (propranolol)
- Glucagon; supportive treatment (Bzd for seizure)

- Calcium channel blockers

- Symptoms: hypotension (e.g., nifedipine)
- Whole bowel irrigation or oral activated charcoal
- Antidote: calcium IV for depressed cardiac contractility

CHEMICAL WEAPONS ATTACK

- Cholinesterase inhibitors

- Insecticides or agents used in terrorist attack (e.g., sarin)
  - common pesticide classes: organophosphates, carbamates

- Toxidrome symptoms: "DUMBELLSS"
  - Stimulation of *muscarinic receptors* causing cholinergic crisis
    - diarrhea, ↑ urination, miosis, bradycardia, emesis, lacrimation, lethargy, salivation, sweating
    - , ↑ bronchial secretion
  - Stimulation of *nicotinic receptors*
    - Hypertension, tachycardia or bradycardia,
    - muscle weakness & bronchospasm, & CNS
    - effects (excitation, confusion, seizures)

- Antidotes: atropine (competitive inhibitor at MR) or
- pralidoxime (act at both MR & NR)

cholinergic crisis: think (more or less) the OPPOSITE of:

Red as a beet
hot as a hare
dry as a bone,
blind as a bat, and
mad as a hatter

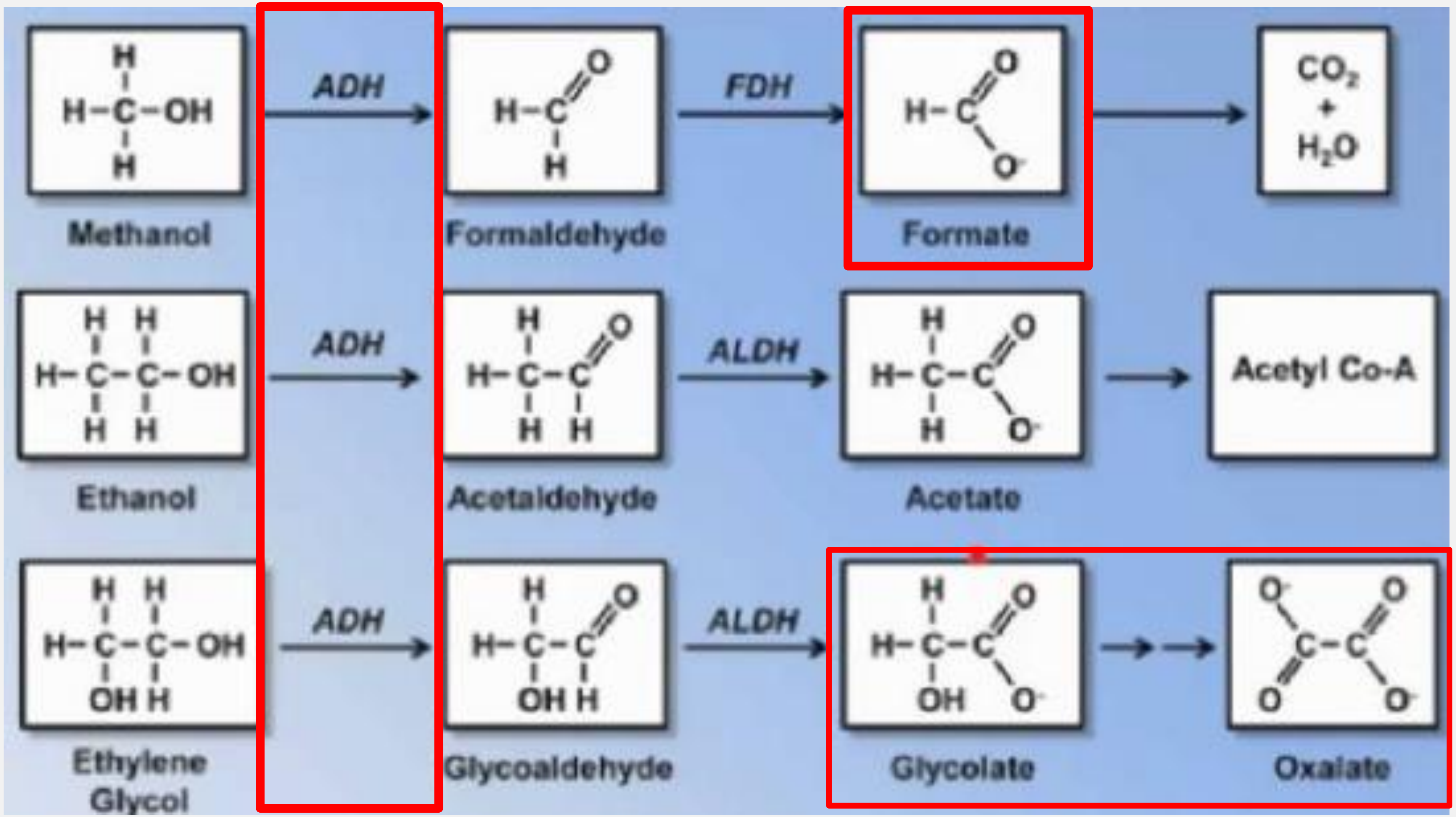
COMMON OVERDOSE/TOXICITY CONT'D

- Digoxin
- Acute toxicity or chronic accumulation (in renal failure or due to drug interactions)
- Symptoms: vomiting, hyperkalemia, cardiac disorders
- Treatments
  - Atropine: effective for AV block or bradycardia
  - Digoxin antibodies (IV): symptoms improve rapidly

- Ethylene glycol & methanol
- Toxic metabolites: toxic organic acids
  - Hippuric acid, oxalic acid, glycolic acids (from EG)
  - Formic acid (from MeOH)
- Symptoms: drunken, CNS depression, high anion gap
- metabolic acidosis, coma or blindness (by formic
- acid), renal failure (by oxalic acid or glycolic acid)
- Antidote: fomepizole (blocks alcohol dehydrogenase) or EtOH.

METABOLISM AND TOXICITY OF VAR. ALCOHOLS

toxic



COMMON OVERDOSE/TOXICITY CONT'D

- Rattlesnake envenomation
- contains destructive enzymes (Phospholipase A2 type) in venom
- Rapid neuromuscular block (sodium channel)
- Symptoms: severe pain, swelling, bruising, hemorrhagic bleb formation, & fang marks
  - Systemic effects
    - Nausea, vomiting, muscle fasciculations, tingling &
    - metallic taste, shock, prolonged clotting time &
    - ↓ platelet count
- Treatment: IV antivenin

- Theophylline
- Accidental overmedication or due to drug interactions ***(e.g., cimetidine, ciprofloxacin, erythromycin)***
- Symptoms: sinus tachycardia, tremor, vomiting are common; β_2-adrenergic effects (hypotension, hypokalemia, hyperglycemia); cardiac arrhythmia, seizures (severe)
- Treatment: activated charcoal/whole bowel irrigation; β blockers; phenobarbital (for seizures), hemodialysis for high conc (>100 mg/L) or if seizure

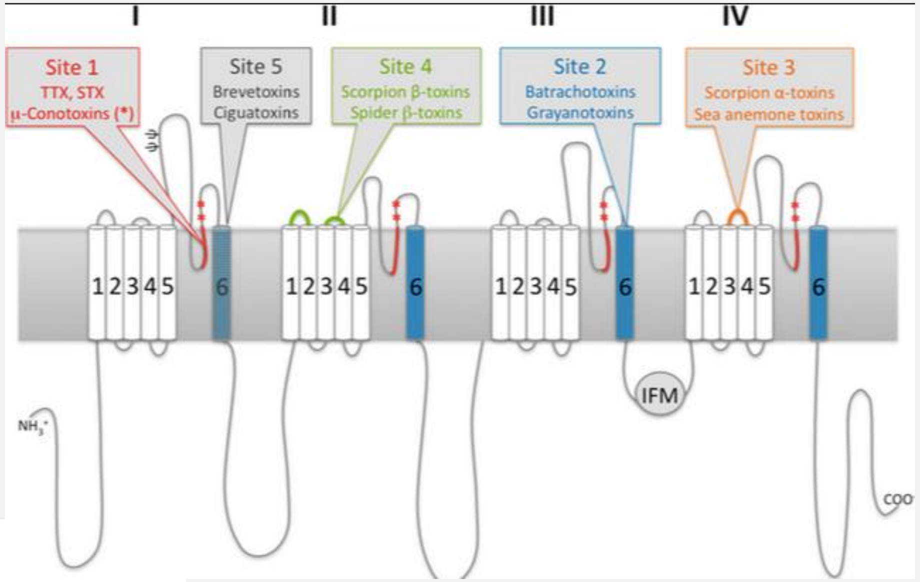
NA$_V$ CHANNEL TOXINS

Na channel:

- scorpion toxins
- puffer fish toxins
- poison frog toxins
- black widow toxins
- Sea anemone toxins

Na, K, Ca channel:

- Cone snail toxins



Site	Neurotoxins	Examples	Peptide	Main binding area known up to date	Result
1	Guanidinium toxins	TTX, STX	–	DI–IV P-loop	Block of Na$^+$ conduction
	μ-Conotoxins	KIIIA, SIIIA, PIIIA	x		
2	Small lipid-soluble toxins	Batrachotoxin	–	DI DIV S6	Negative shift in voltage-dependency of activation
		Veratridine	–		Slowing down of inactivation
		Grayanotoxins	–		Block of Na$^+$ conductance
		Aconitine	–		Altering ion selectivity
3	Scorpion α-toxins	AaH II, LqhaIT, BMK M1	x	DIV S3–S4	Slowing down of inactivation
	Sea anemone toxins	ATX-II, AFTII	x		
4	Scorpion β-toxins	Css4, Tsγ, AahIT	x	DII S3–S4	Negative shift in voltage-dependency of activation
	Spider β-toxins	Magi 5, HWTX-IV	x		Block of Na$^+$ conductance
	μO-conotoxins	MrVIA	x		
5	Cyclic polyether compounds	Brevetoxins	–	DI S6	Negative shift in voltage-dependency of activation
		Ciguatoxins	–		Slowing down of inactivation
6	δ-Conotoxins		x	DIV S4	Slowing down of inactivation
7	Pyrethroids	DDT, Deltamethrin	–	DII–DIII	Slowing down of inactivation
LA	Local anesthetics	Lidocaine	–	DIV S6	Block of Na$^+$ conduction
	Anticonvulsants		–		
	Antiarrhythmics		–		
	Antidepressants		–		

Treatment: antivenom if available, supportive care

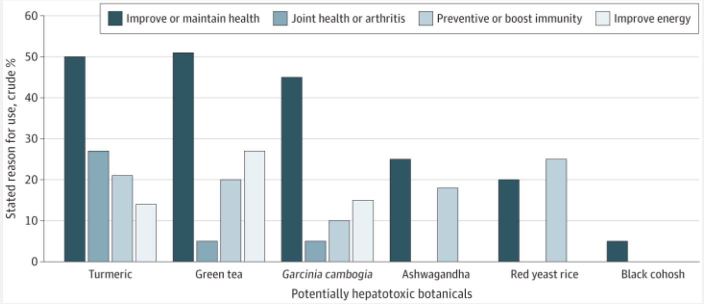
take-home: don't memorize! know common target (NaV) but that different binding sites exist

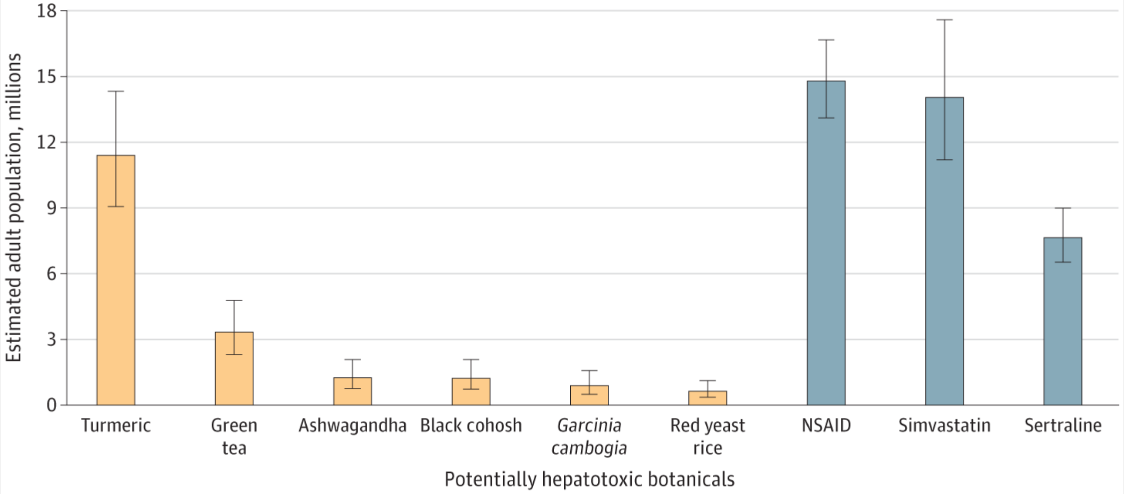
NATURAL/DIETARY SUPPLEMENTS

- % drug-induced liver tox cause by dietary supplements:
  - 2004/5: 7%
  - 2013/14: 20%
    - source: https://dilin.org/

- Treatment: discontinue supplement use, supportive care
  - https://www.uptodate.com/contents/hepatotoxicity-due-to-herbal-medications-and-dietary-supplements#H734050788

reasons for using hepatotoxic supplements:





how supplement use compared to hepatotoxic medication use

https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2821951

QUICK REVIEW: COMMON TOXINS AND TREATMENTS

- How does NAC work as a treatment? What toxic exposure does it treat?

  acetaminophen overdose

- What is the role of bicarbonate infusion in treating aspirin toxicity?

  increasing buffer to negate the salicylate anion and decrease anion gap

- What is the role of ethanol in treating methanol toxicity?

  take up ADH enzyme so less toxic products are made

QUICK REVIEW: COMMON TOXINS AND TREATMENTS ANSWERS

- How does NAC work as a treatment? What toxic exposure does it treat?
  - N-acetyl cysteine (NAC) treats APAP toxicity. Glutathione (GSH) is an endogenous antioxidant that conjugates and inactivates the toxic APAP metabolite NAPQI. GSH is depleted by high NAPQI levels that result from APAP overdose. NAC provides cysteine as a component precursor to support glutathione synthesis (as GSH is a tripeptide of Cys, Gly, Glu)

- What is the role of bicarbonate infusion in treating aspirin toxicity?
  - accepted mechanism: it helps to alkalinize urine, shifting salicylic acid toward its deprotonated (and ionized) state, "trapping" it in the bladder and substantially (up to 10X) increasing its excretion rate
    - but FYI some debate of this assertion; acid-base distribution suggests that this can't be the only mechanism: https://pubmed.ncbi.nlm.nih.gov/15181662/

- What is the role of ethanol in treating methanol toxicity?
  - it competes with methanol at the alcohol dehydrogenase enzyme, slowing methanol oxidation and to formahdehyde (and then formate)

LECTURE OUTLINE

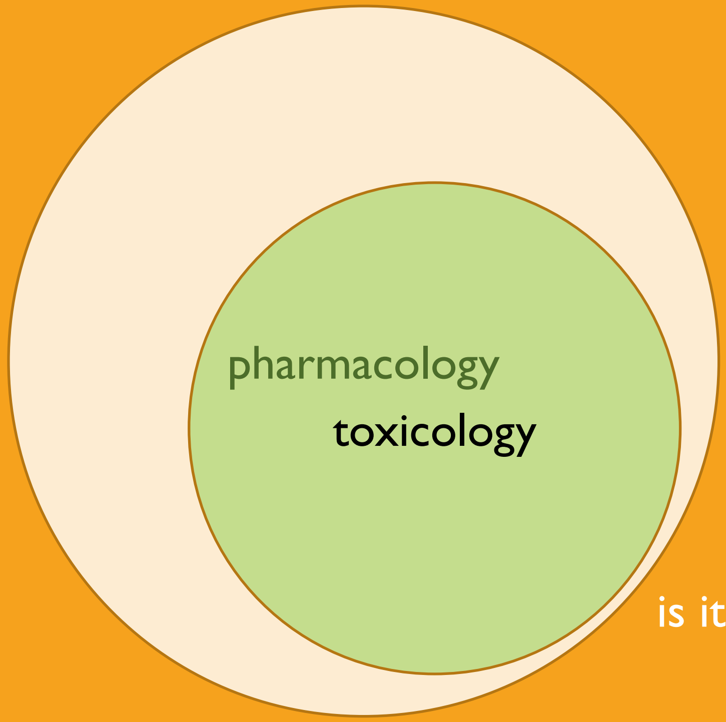
Drug lists (slides 4-5: on your own)

Drug toxicity: managing the poisoned patient (Katzung 58, Goldfrank's)

Individual common toxins (on your own)

Occupational Tox: Common air and envt pollutants (Katzung 56)

Heavy metals (Katzung 57)



is it a little cramped in here?

OCC/ENVT TOX TERMS AND DEFINITIONS TO KNOW

- **Occupational toxicology**: effects of chemicals or compounds found in the workplace, on workers exposed to them
- Agency: OSHA (formerly BLS)
- Goals
  - identify agents of concern and acute and chronic diseases caused
  - treat these diseases
  - surveillance of exposed workers
  - establishing recommended exposure limits
    - Permissible Exposure Limit (PELs): **legal workplace limits**
      - see www.osha.gov , www.msha.gov
    - Threshold Limit Values (TLVs): **recommended product and exposure limit guidelines**
      - see https://www.osha.gov/annotated-pels/note

- **Environmental toxicology** ("ecotoxicology"): effects of chemicals on envt and ecology
- Agency: EPA, FDA
- Goals
  - identify pathways of exposure/accumulation
  - toxicology of individual organisms
  - impact on populations/species within envt
    - humans too, urban/developed envts as well as "naturally occurring"
  - climate change
  - global cross-contamination water and air usually

TERMS AND DEFINITIONS TO KNOW CONT'D

- **Hazard**: ability of a chemical agent to cause injury in a given situation or setting
  - what matters here? CONDITIONS of use and EXPOSURE predicted based on conditions
- **Risk**: the expected frequency of an undesirable effect arising from exposure to a chemical or physical agent
  - what matters here? HAZARD times EXPOSURE (quantity, duration, intensity)
- Primary routes of exposure
  - inhalation
  - skin contact

TERMS AND DEFINITIONS TO KNOW CONT'D

- **Bioaccumulation**
  - envt (more accurately biomagnification): increasing concentrations of toxins by "higher" levels of food chain
  - patient: time-dependent accumulation of slowly degraded/eliminated toxins
- **Toxidrome**: toxic + syndrome: collection of symptoms (syndrome) caused by toxin exposure/accumulation
- Note: be careful in pt counseling to use "toxin" correctly- understand how clinical use differs from colloquial use and be able to explain the difference to pt in clear language
  - e.g., a "detox diet" is NOT what we are addressing here
    - also not a real thing

FYI: biomagnification in the Great Lakes biome

Source	PCB Concentration (ppm)[1]	Concentration Relative to Phytoplankton
Phytoplankton	0.0025	1
Zooplankton	0.123	49.2
Rainbow smelt	1.04	416
Lake trout	4.83	1932
Herring gull	124	49,600

Katzung

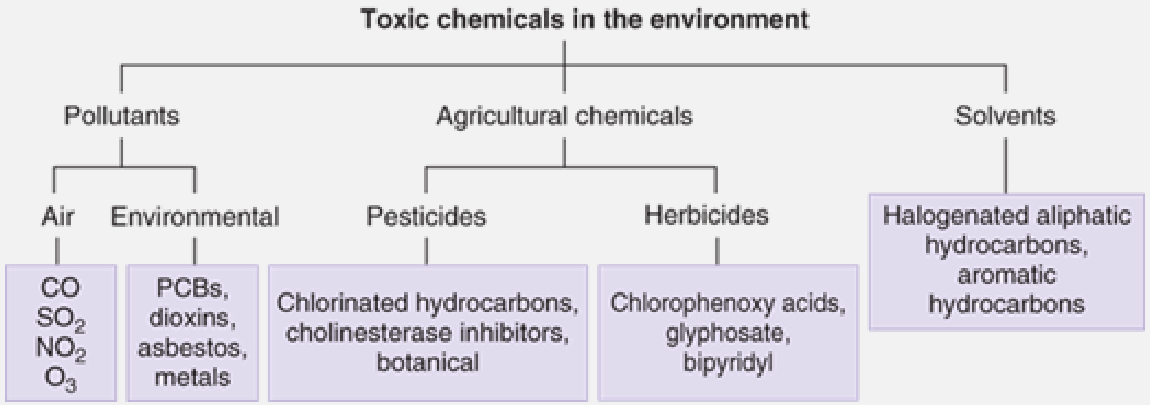
COMMON ENVT TOXINS AND WHERE TO FIND THEM

- Air pollutants: widely dispersed
  - **carbon monoxide (CO)**
    - fuel-burning engines, heat sources
  - **sulfur dioxide (SO_2)**
    - fossil fuel combustion
  - **nitrogen oxides (esp NO_2)**
    - but not nitrous oxide (N_2O)!
    - diesel/auto emissions, agriculture
  - **ozone (O_3)**
    - high-voltage equipment, fuel burning
- Air pollutants: occupational exp.
  - aromatic and halogenated hydrocarbons as manufacturing solvents:
    - **benzene, chloroform, toluene, trichloroethane, trichloroethylene**
  - as refrigerants/propellants:
    - **carbon tetrachloride (CCl_4)**
  - in drycleaning:
    - **tetrachloroethylene**

KNOW

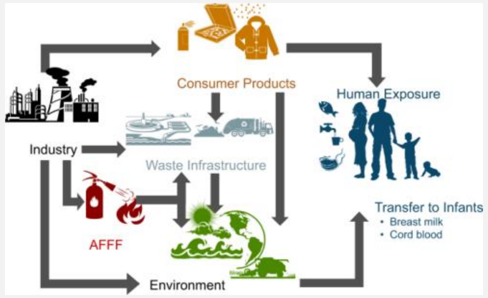
- Environmental pollutants
  - **polychlorinated biphenyls (PCBs)**
    - engine/electrical coolants and lubricants
    - use banned in 1979
  - **per/poly-fluoroalkyl substances (PFAS)**
    - FYI nomenclature note: formerly PFCs, PFOAs
    - non-sticks, water repellants, flame-retardants
  - endocrine disruptors
    - plastics, flushed medications, plant-derived, others
- cyanotoxins
  - blue-green algal blooms
- particles
  - asbestos
  - UFPs (nanoparticles)

- Herbicides
  - chlorophenoxy-type
    - e.g., **glyphosate, Agent Orange**
  - bipyridals
    - e.g., **paraquat**
- Pesticides/insecticides
  - organochlorines
    - e.g., **DDT**
  - organophosphates
    - e.g., **malathion**
  - carbamates
  - plant-derived
    - e.g., **nicotine, rotenone, pyrethrum**

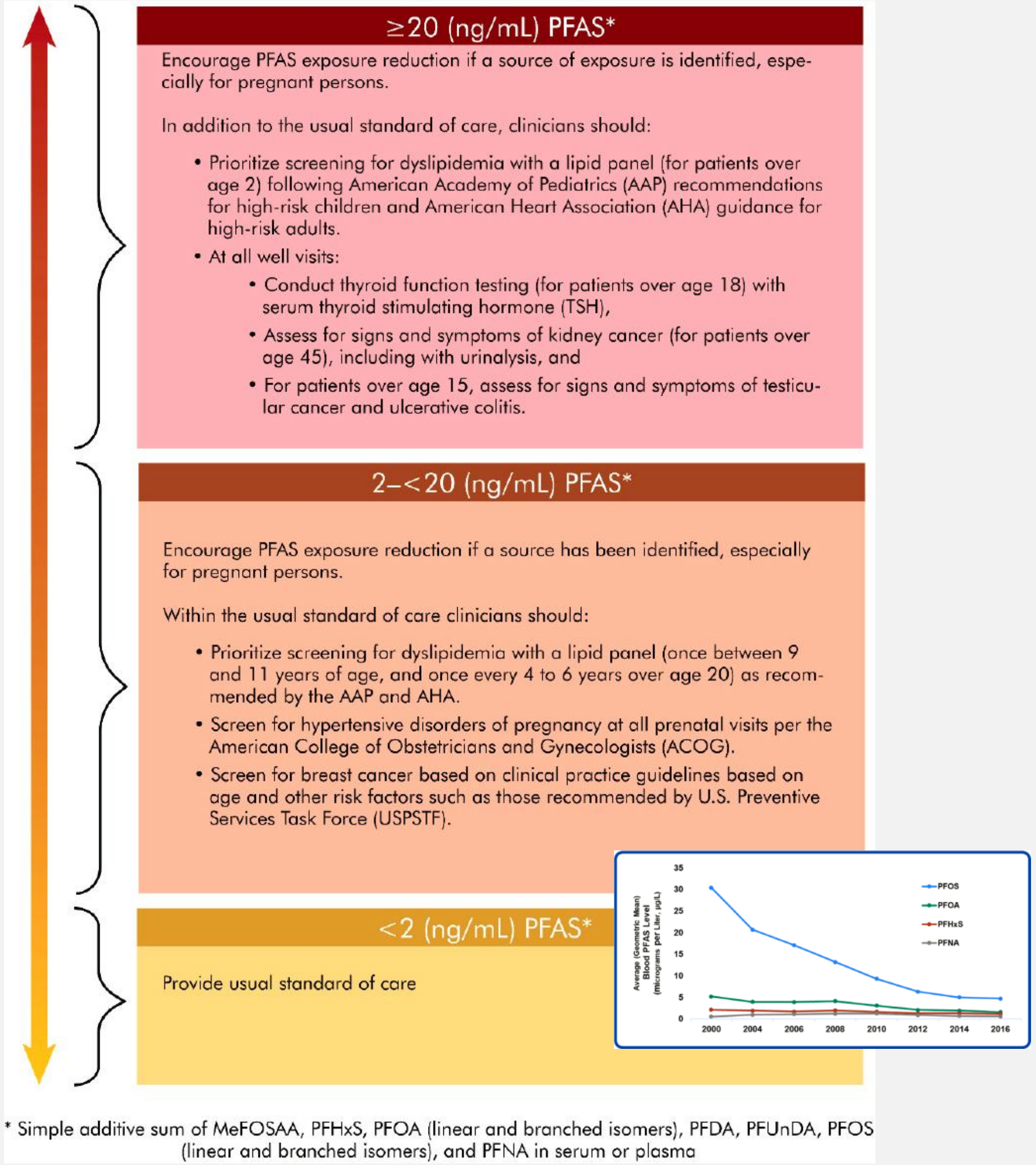


COMMON TOXINS EMERGING EXAMPLE: PFAS

- aka perfluorochemicals
  - found in: industrial production, non-stick cookware, fire retardants, stain retardants
- systems affected by exposure
  - immune: decreased antibody response, increased risk of kidney cancers
  - metabolism: dyslipidemia
  - reproductive: some evidence for increased risk of gestational hypertension
  - developmental: delays in fetal/neonatal growth



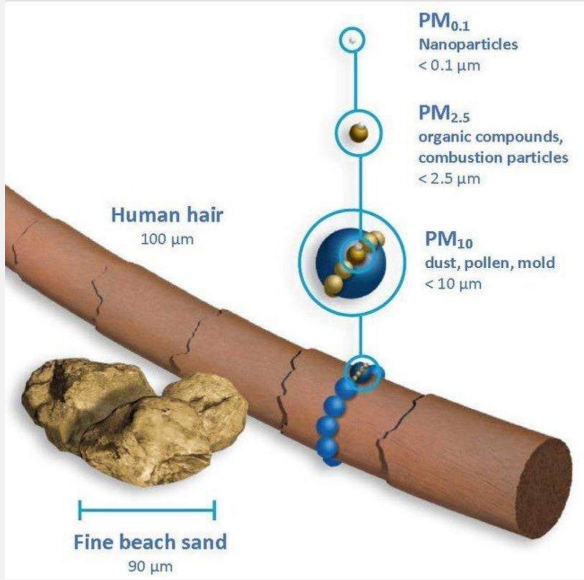
- Treatments
  - no current treatments
- clinical approach
  - counseling on exposure reduction
    - e.g., water filtration
  - for significant exposure, testing recommendations:



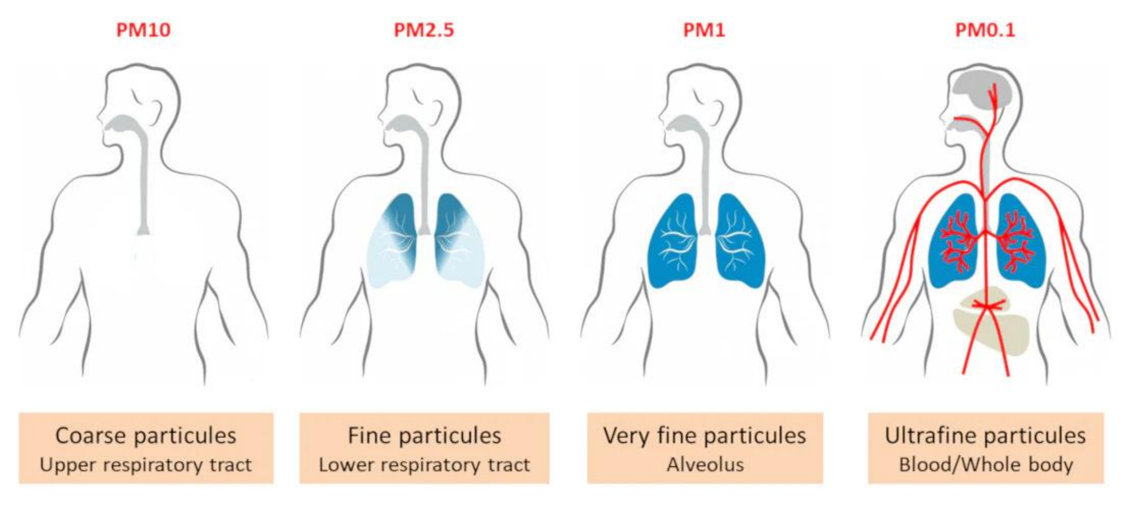
https://nap.nationalacademies.org/read/26156/chapter/2#13

COMMON TOXINS EMERGING EXAMPLE: UFP

- ultra-fine particle (UFP): <0.1µM
  - unlike larger particles, UFPs are MORE LIKELY to:
    - remain suspended
    - enter circulation
- composition: varied, typically metals, hydrocarbons, sulfates
- common sources:
  - combustion (industrial, vehicular, wildfires)
  - concrete/construction
  - tire and brake wear
- toxicity: inflammatory/oxidative-stress
  - respiratory and CVD
  - metals exposure
  - carcinogenic



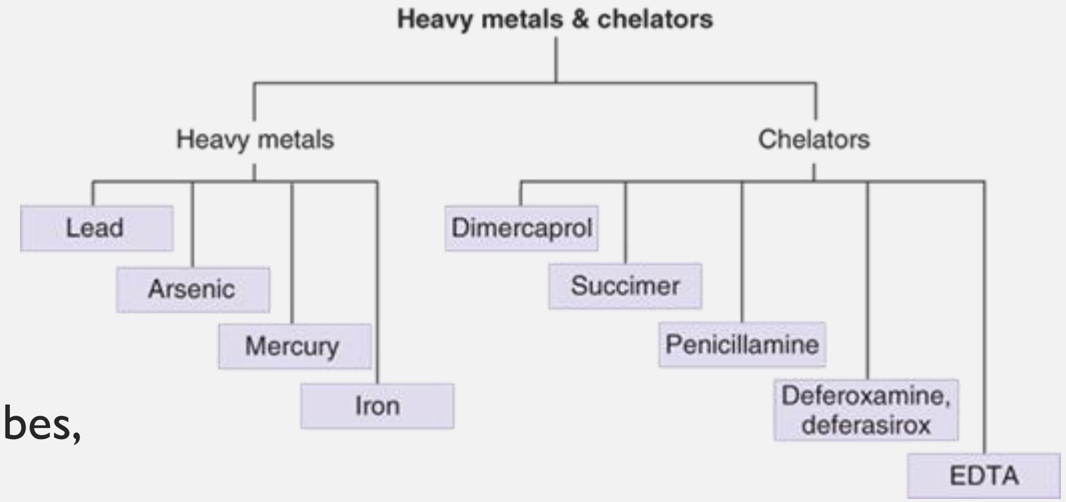
https://phys.org/news/2022-02-ultrafine-particle-sensors-drastic-air.html



COMMON TOXINS* AND WHERE TO FIND THEM: METALS

- **beryllium (Be)**
  - electronics, ceramics, dental equipment
- **cadmium (Cd)**
  - paints/pigments, batteries, semiconductors, plastics
- **lead (Pb)**
  - paints/pigments, metal alloys, ceramics, ammunition
- **arsenic (As)**
  - paints/pigments, semiconductors, wood preservatives, herbicides, drinking water, foods grown with As-contaminated groundwater
    - apples, rice
- **mercury (Hg)**
  - paints/pigments, mining, older electronics, fluorescent light bulb/tubes, older dental amalgam

FYI here, from Dr. Nguyen's ODS slides Fall 2021



KNOW

*not a complete list

MOLECULAR MECHANISMS OF TOXINS COMMON THEMES

ACUTE

- mitochondrial toxins
  - e.g., ETC complex inhibitors
- cholinergic signaling derangement
  - anticholinergics
  - cholinesterase inhibition/cholinomimetics
    - e.g., organophosphates, carbamates
- oxidant/antioxidant disruptors
  - e.g., paraquat
- cation disruptors
  - e.g., lead

CHRONIC

- carcinogens
  - e.g., radium, soot
- respiratory irritants/accumulants
  - e.g., asbestos, nanoparticles
- endocrine disruptors
- neurotoxins

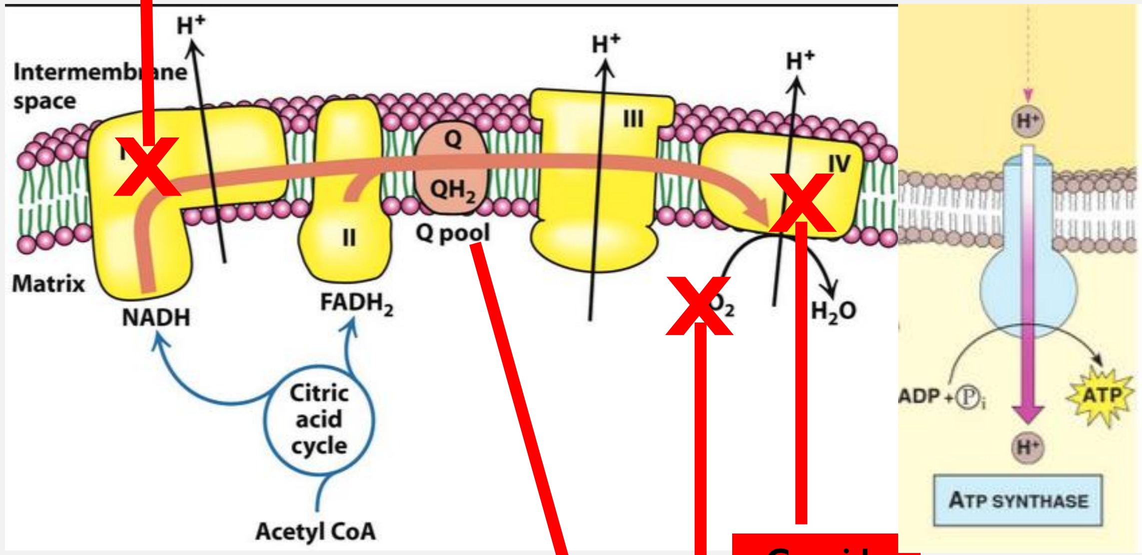
EXAMPLE ACUTE MECHANISM: MITOCHONDRIAL TOXINS

organic pesticide

Rotenone

ETC generates protonmotive force

ATP synthase consumes pmf to make ATP



Cyanide
Azide

specific CIV inhibitors

Oxidative phosphorylation: a 3-step energy transduction mechanism.

1. substrate oxidation yields e-
2. e- are carried to ETC, energy transduced to protonmotive force (pmf)
3. pmf is used to power ATP synthesis

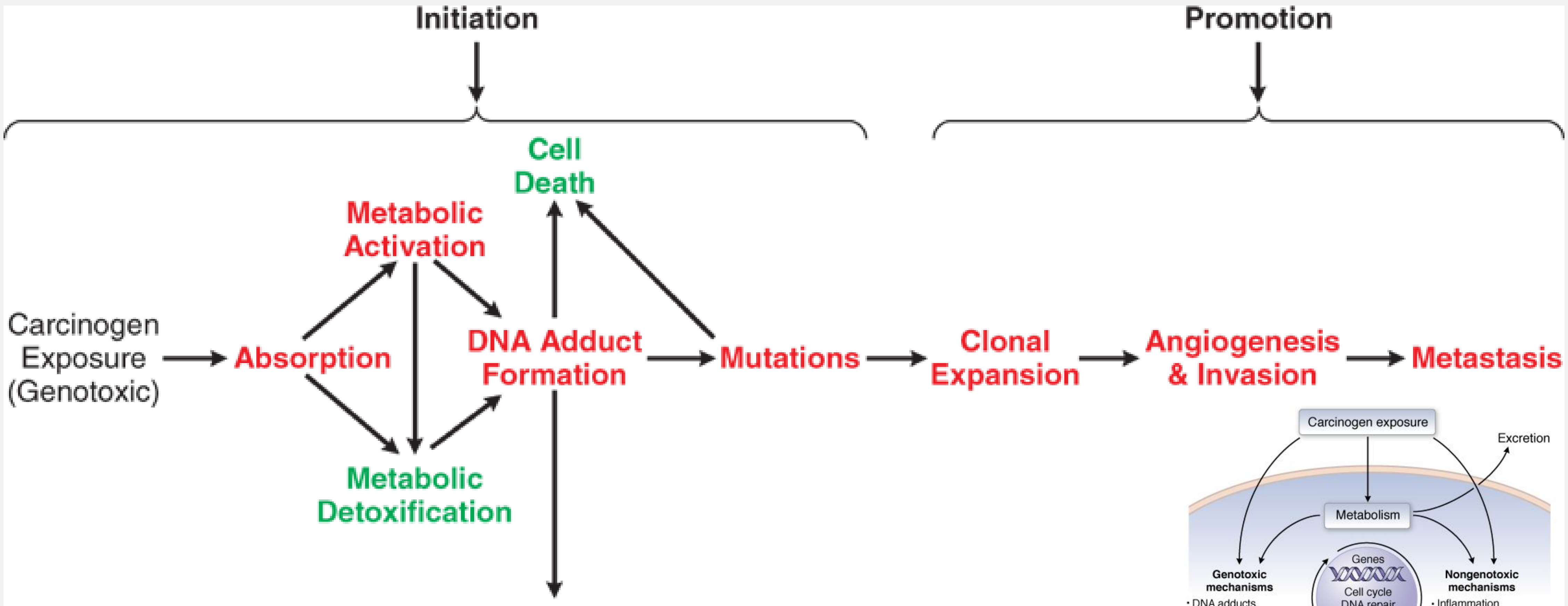
CO

decreases O_2 availability

paraquat

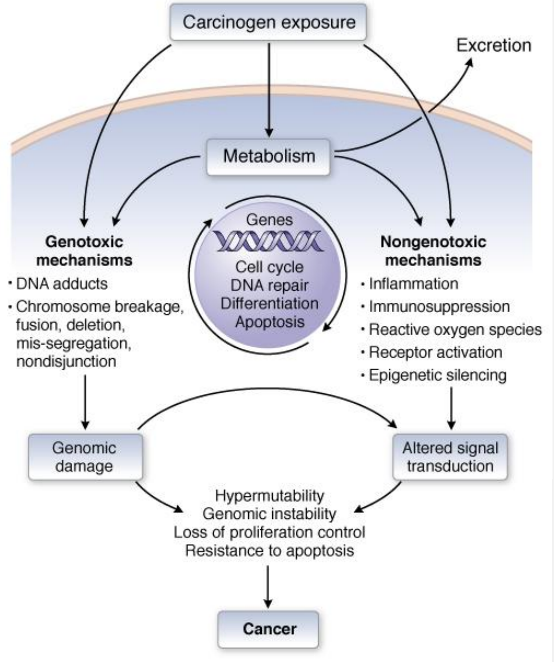
ROS production, overall mitochondrial collapse

EXAMPLE CHRONIC MECHANISM: CARCINOGENS



Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition:

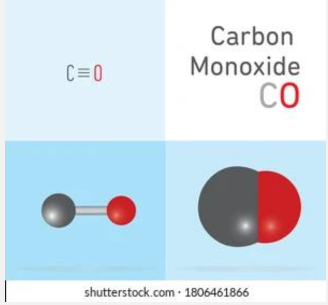
note: this illustrates genotoxic mechanisns; non-genotoxic also possible:



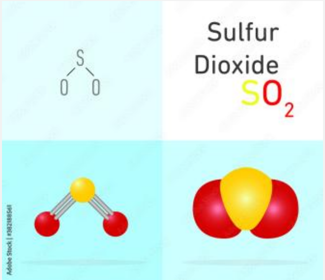
FYI for toxicology

COMMON AIR POLLUTANTS

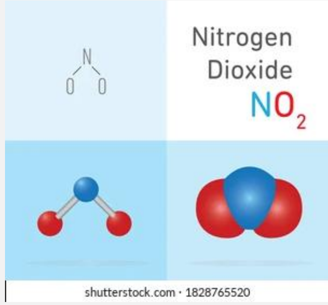
- Carbon monoxide (CO)
  - A byproduct of incomplete combustion
  - Colorless, tasteless, odorless, non-irritating gas



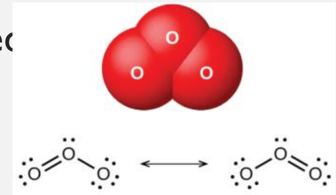
- Sulfur dioxide (SO_2)
  - Generated by the combustion of sulfur-containing fossil fuels
  - Colorless irritant gas



- Nitrogen oxides (NO, NO_2)
  - NO_2: Brownish irritant gas sometimes associated with fires
  - accumulates in agricultural settings, e.g., NO_2 in silos



- Ozone (O_3)
  - Bluish irritant gas: protective in troposphere (no real exposure)
  - irritant exposure at ground level: occurs around high-voltage electrical equipment & around devices used for air/water purification, as well as polluted urban air (Nitrogen oxides + VOCs → O_3)

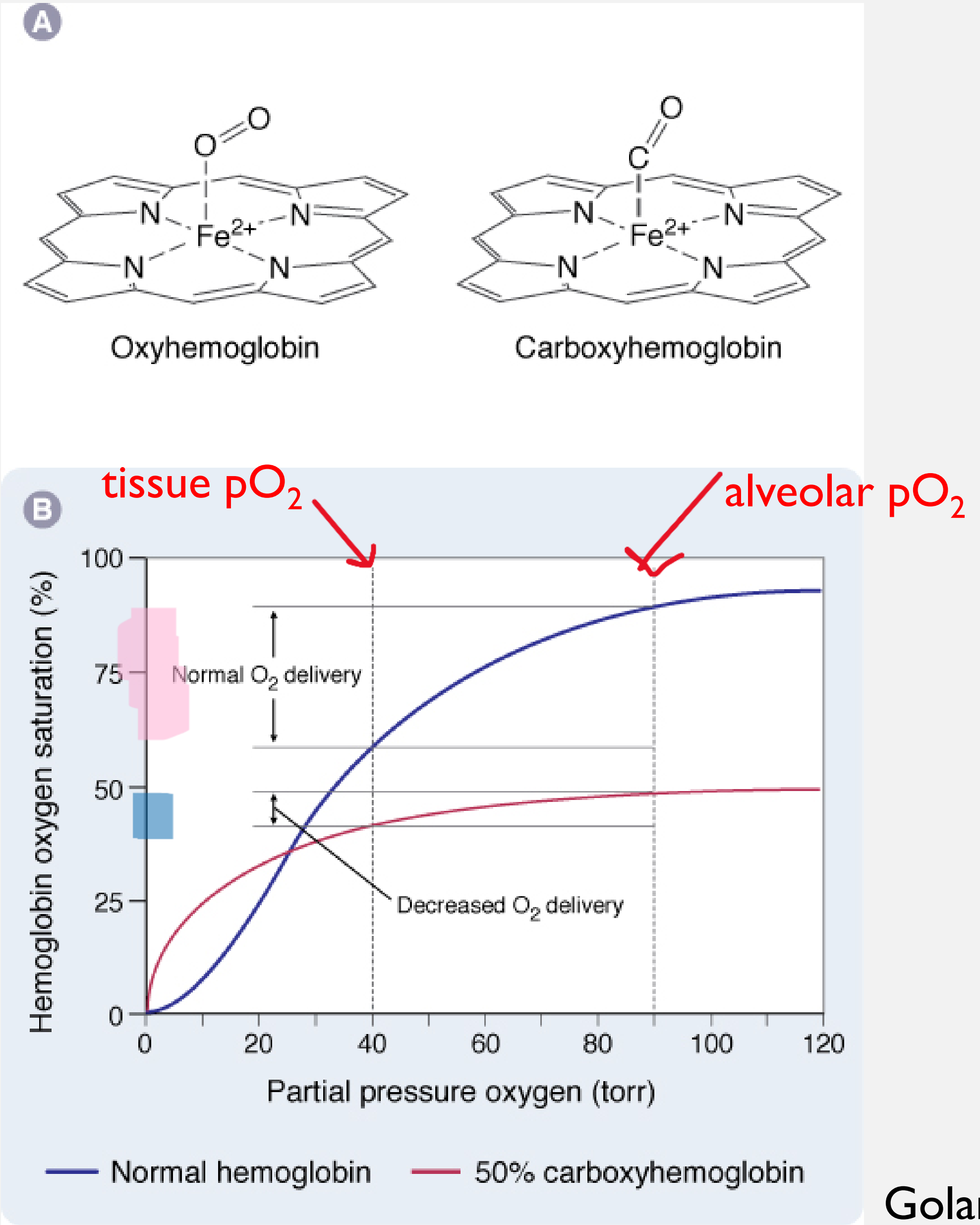


CARBON MONOXIDE

- Where found? ANY combustion source
  - but common unintended exp through:
    - car exhaust, propane heaters/generators/camp stoves
    - gas ranges, wood stoves
    - cigarette/other smoking
    - house fires

- High affinity to Hb (>200x higher than O_2)
- Carboxyhemoglobin *decreases* O_2 binding to Hb (also other heme-containing proteins: myoglobin, cytochromes)
- most sensitive tissues: brain, heart

- Exposure correlates:
  - >3% CO-Hb (normal <1%); for smokers, >10% (normal up to 5-10%): unusual exposure
  - >30-40%: headache, vomiting, visual disturbances
  - >50-60%: seizure, convulsions
  - >70% death likely



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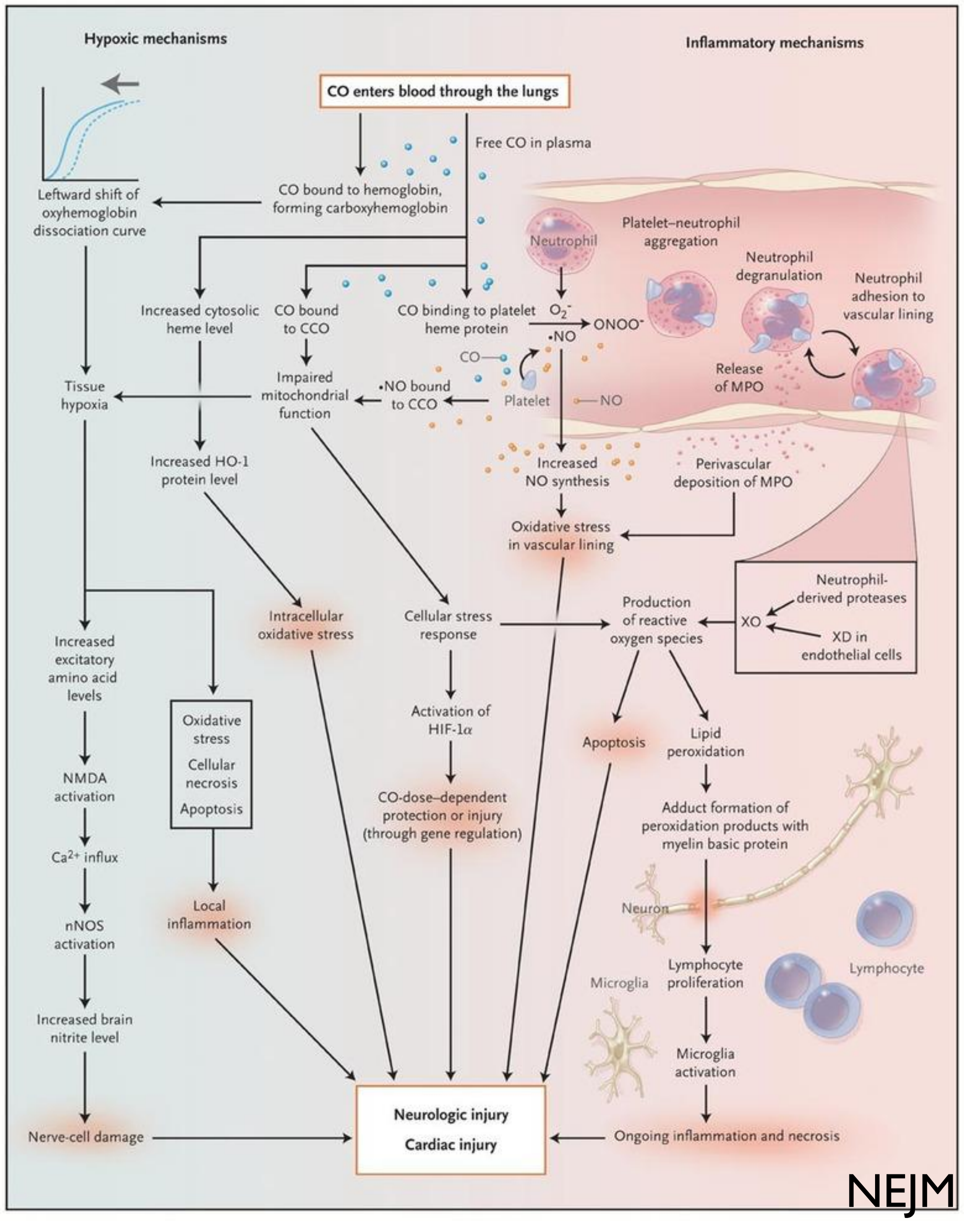
MOA AND TREATMENT NOTES

- Carbon monoxide (CO) symptoms:
  - Psychomotor impairment
  - Headache & tightness in the temporal area
  - Confusion & loss of visual acuity
  - Tachycardia, tachypnea, syncope & coma (> 40% COHgb)
  - Deep coma, convulsions, shock & respiratory failure
  - Prolonged hypoxia may cause *irreversible* brain damage
  - Carboxyhemoglobin > 60% may cause death
  - long-term: increased mortality/morbidity risk
    - esp in CV-compromised pt
- Treatment:
  - removal from CO exposure, normo- or hyper-baric oxygen



O_2 WILL EVENTUALLY REPLACE CO, BUT…..

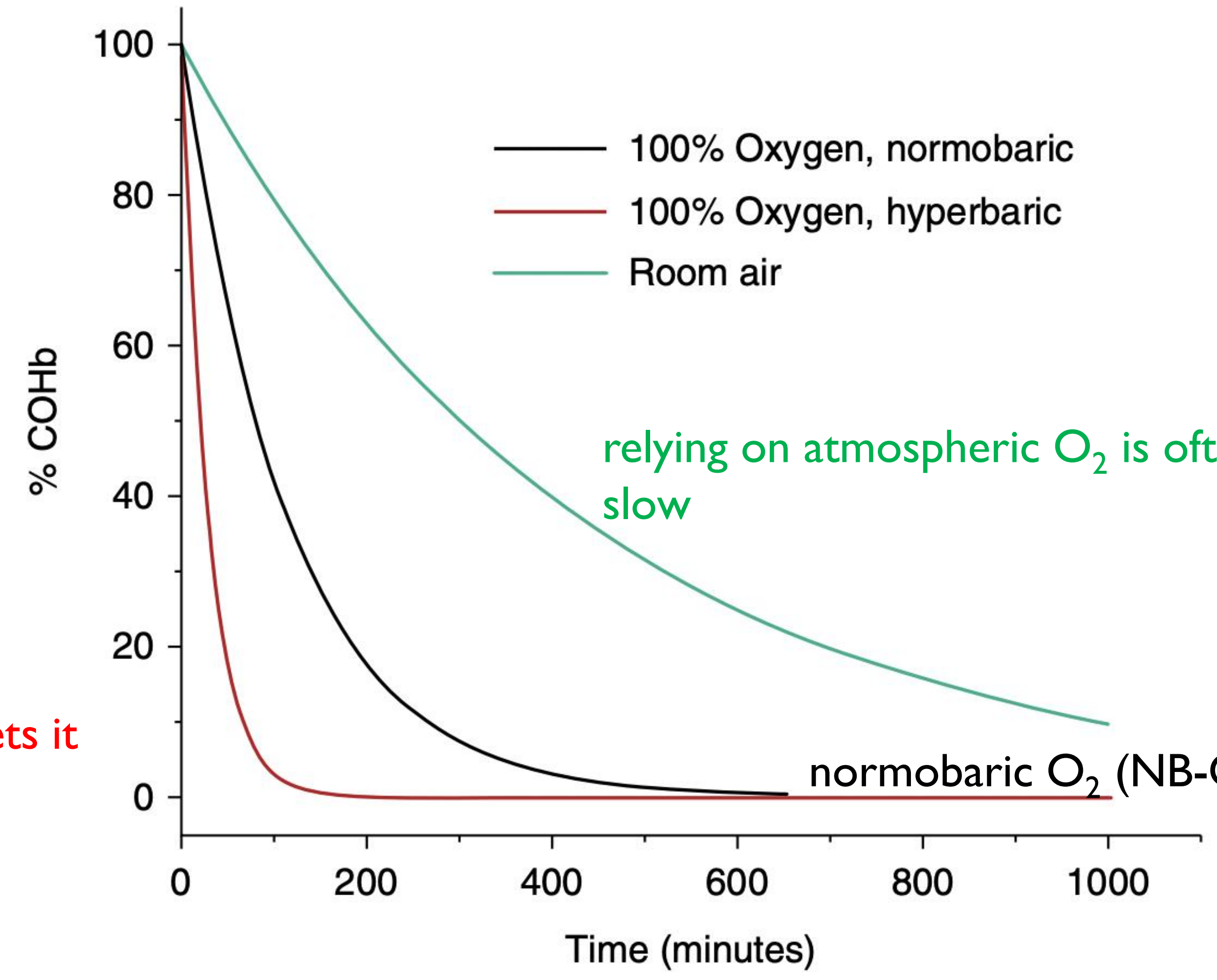


Figure 3. Decay of the carboxyhemoglobin (COHb) species under therapeutic treatments. Half-life values of COHb in room air (320 min), 100% normobaric oxygen (74 min), and 100% hyperbaric oxygen (HBO_2; 20 min) determined from Refs. 12, 74, 75, 77–79.

relying on atmospheric O_2 is often too slow

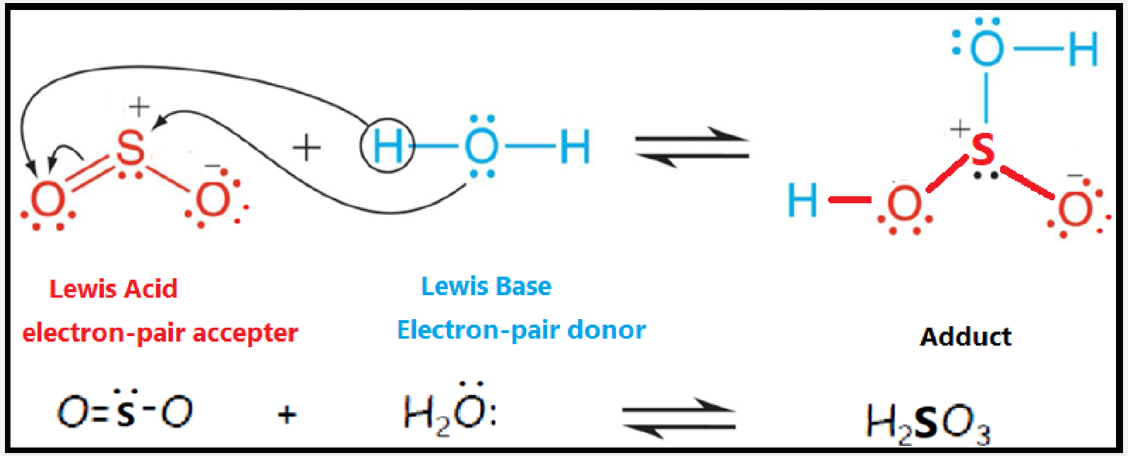
hyperbaric O_2 (HB-O_2) gets it done faster

normobaric O_2 (NB-O_2) is usually fast enough

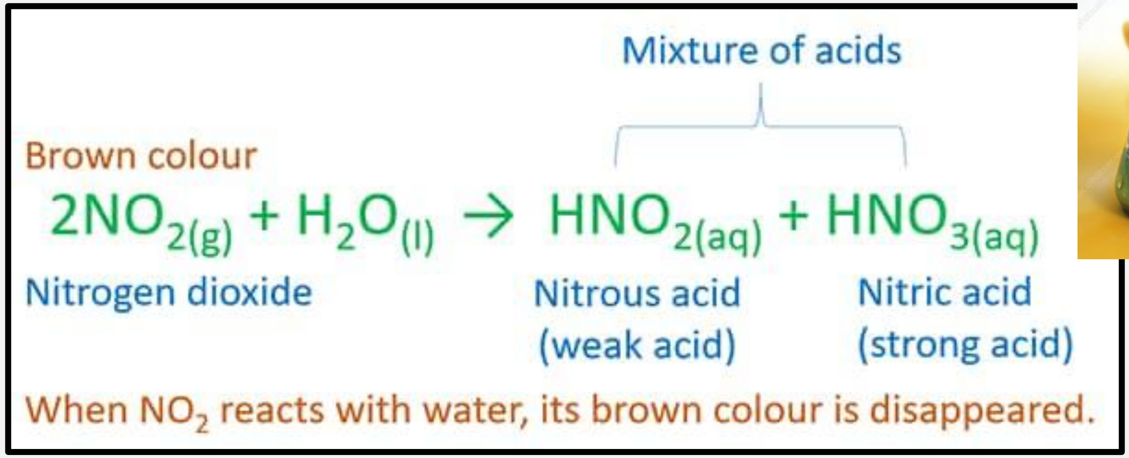
DOI: 10.1164/rccm.201606-1275CI

SULFUR AND NITROGEN OXIDES

- Sulfur dioxide (SO_2)
  - from e.g., coal/diesel burning, paper mfr., metal smelting
  - Forms sulfurous acid (H_2SO_3) on contact with mucous membranes: irritant effect
    - Inhaled SO_2 can cause bronchospasm
  - Treatment: same as for irritation of the respiratory tract: oxygenation & alveolar ventilation; bronchodilators

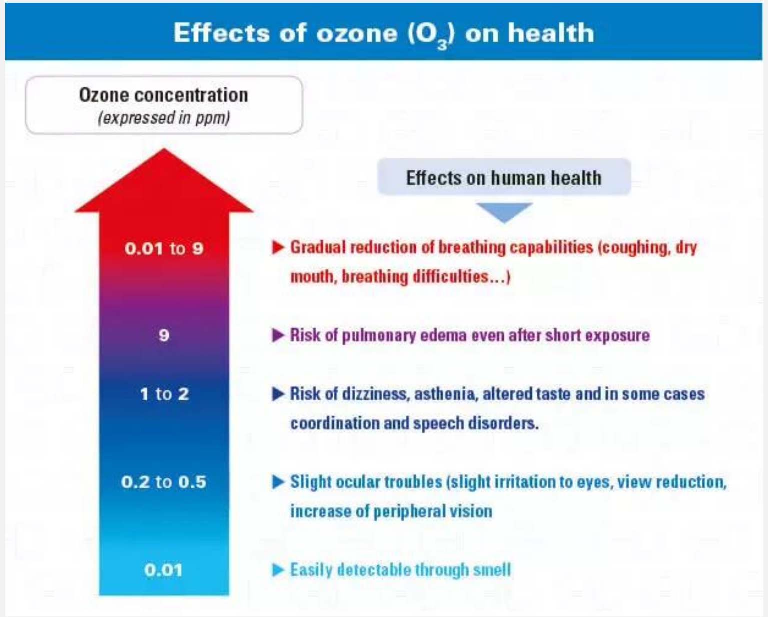


- Nitrogen oxides (NO, NO_2)
  - Produces pulmonary edema; irritation, cough, mucoid sputum, dyspnea, chest pain (Silo-filler's disease)
  - Treatment: oxygenation & alveolar ventilation; bronchodilators



GROUND-LEVEL OZONE, SOLVENTS

- Irritant for mucous membranes: produces URT irritation or deep lung irritation with pulmonary edema (severe)
- Shallow, rapid breathing; decrease in pulmonary compliance; ↑ sensitivity to bronchoconstrictors
- Management: similar to NO_2 or SO_2 exposure

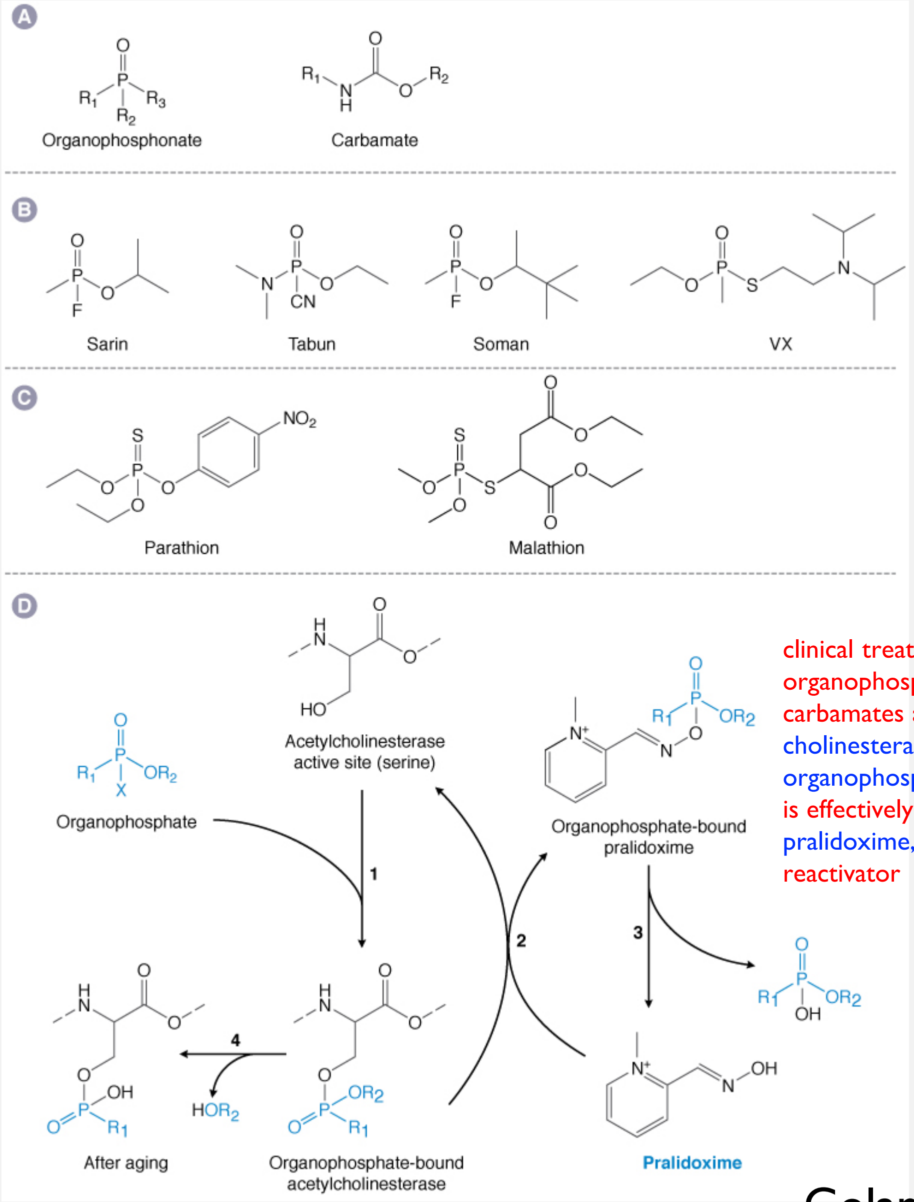


- Solvents
- Halogenated aliphatic hydrocarbons
  - Chloroform, trichloroethane, trichloroethylene, etc
  - Potentially carcinogenic (e.g., trichloroethylene)
  - CNS depression, liver, kidney, cardiac toxicity
  - Symptomatic management
- Aromatic hydrocarbons
  - Benzene: CNS depression, myelotoxicity (e.g., leukemia)
  - Toluene: CNS depression

TYPE	EXAMPLE	USE	PATHOPHYSIOLOGY	COMMENTS
Aliphatic petroleum distillates	Methane Propane Butane Gasoline Kerosene Mineral spirits Mineral oil Naphtha Mineral seal oil Diesel oil *n*-Hexane	Fuels Liquid fuels Solvents Furniture polish Degreasers Multiple uses in chemical industry	Asphyxiants causing hypoxia and CNS depression Abused inhalants Pneumonitis when aspirated CNS depression from fumes *n*-Hexane causes peripheral neuropathy	Sudden death from intentional inhalation Viscosity and volatility determine spectrum of toxicity Mineral seal oil has high aspiration potential Poor gastrointestinal absorption
Aromatic petroleum distillates	Toluene Xylene Benzene	Used in plastics, pharmaceutical, rubber, chemical, and solvent industries Degreasers	Highly volatile, lung aspiration Absorbed from gastrointestinal tract Abused inhalants	Inhaled toluene causes renal tubular acidosis Benzene causes aplastic anemia, leukemia
Wood distillates	Turpentine Pine oil	Solvent Household disinfectants	Well absorbed from gastrointestinal tract	Gastrointestinal and CNS toxicity
Halogenated hydrocarbons	Methylene chloride Chloroform Carbon tetrachloride Trichloroethylene Freon Methylbromide Lindane DDT	Solvents Cleaning fluids Degreasers Fire extinguishers Paint strippers Fumigants	Multisystem toxicity (CNS, renal, hepatic, cardiac) Inhalant abuse Highly lipid soluble	Methylene chloride metabolized to carbon monoxide Carbon tetrachloride is radiopaque Insecticides absorbed through skin
Related chemicals	Phenol Creosols	Disinfectants	Very corrosive	Phenol causes severe skin burns

PESTICIDES AND HERBICIDES

- Insecticides
  - Organochlorine insecticides e.g., DDT
  - Organophosphorus insecticides e.g., malathion
  - Carbamate insecticides, e.g., aldicarb
  - Plant-based/botanical insecticides e.g., rotenone
  - pyrethroid insecticides, e.g., permethrin
    - bind Na_V channels → delayed AP termination
- Herbicides
  - Chlorophenoxy herbicides e.g., glyphosate (Round-Up)
  - Bipyridyl herbicides e.g., paraquat



clinical treatment note: organophosphates and carbamates are both anti-cholinesterases, but only organophosphate exposure is effectively treated with pralidoxime, a cholinesterase reactivator

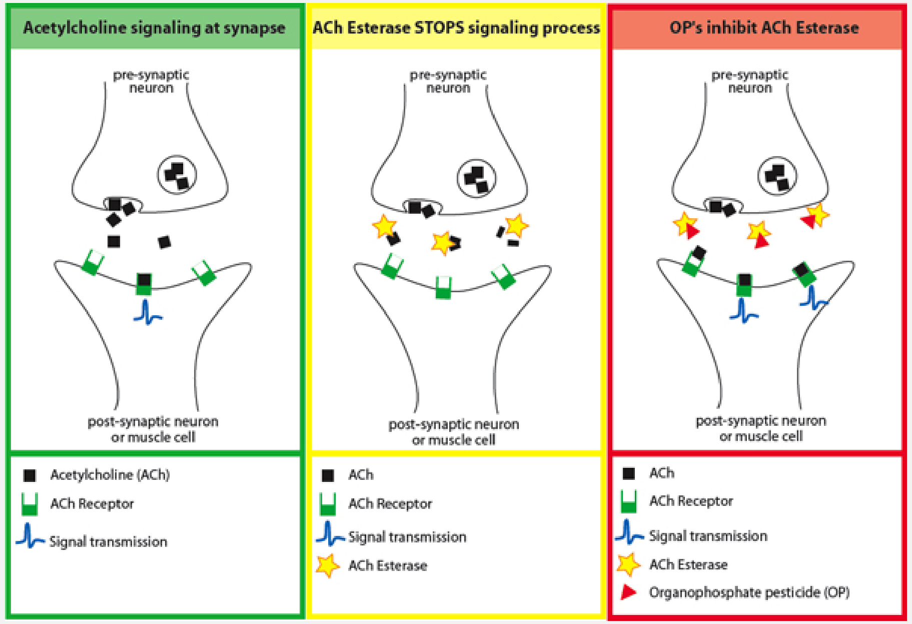
Golan

PESTICIDES

- **Organochlorine** (**DDT**, benzene hexachloride, cyclodienes,
  - toxaphenes**):** *persistent pesticides*
  - *Slow* degradation (environmental concern)
  - Cause rapid repetitive firing in most neurons
  - (by interfering with inactivation of the sodium channels
  - in excitable membranes)
  - **CNS stimulation** (tremor, convulsion), endocrine dysfunction
  - Symptomatic management

- **Organophosphorus (e.g., malathion)**
  - **Inhibit acetylcholinesterase through *phosphorylation***
  - Some also phosphorylate neuropathy target *esterase (NTE)*
  - Less environmental impact (rapid biotransformation)
  - **Symptoms: cholinergic effects; delayed polyneuropathy**
  - (OPIDP; burning/tingling sensation, motor weakness)
  - **Management: Atropine** and **Pralidoxime**

reverses inhibition of AChE



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PESTICIDES CONT'D

- **Carbamate insecticides**
  - **Inhibit acetylcholinesterase by *carbamoylation***
  - Symptoms: shorter duration than organophosphorus
    - FYI efficacy of pralidoxime as treatment for carbamate tox debated: rec by uptodate and Goldfrank's, but not by Katzung)

- **Plant-derived ("botanical") insecticides**
  - **Nicotine**
    - Rapidly absorbed from mucosal surface
    - Binds to nicotinic receptors of the postsynaptic
    - membrane, causing *depolarization*
    - Treatment: maintain vital signs & suppress convulsion
  - **Rotenone**
    - Inhibits Complex I of the mitochondrial electron transport chain
    - GI irritation, conjunctivitis, dermatitis, rhinitis
    - Symptomatic treatment
  - **Pyrethrin**
      - CNS stimulation (convulsion), contact dermatitis
    - Symptomatic treatment

HERBICIDES ETC.

- Chlorophenoxy herbicides
  - 2,4-dichlorophenoxyacetic acid (2,4-D)
  - 2,4,5-trichlorophenoxyacetic acid (2,4,5-T)
  - Uncouple oxidative phosphorylation
  - Symptoms: coma, generalized muscle hypotonia
  - Treatment: Hemodialysis

- Bipyridyl herbicides
  - Paraquat (a quaternary ammonium herbicide)
  - Produce *free radicals*; accumulate in *lung*
  - Symptoms: GI irritation (hematemesis, bloody stools)
    - respiratory distress, hemorrhagic pulmonary edema
  - Treatment: symptom-based, gastric lavage

- Polychlorinated biphenyls (PCBs, coplanar biphenyls)
  - Industrial use terminated in 1977 in USA
  - Highly stable and lipophilic; resistant to degradation
  - *Bioaccumulates in food chains* (e.g., fish)
  - Treatment: symptom-based

- Polychlorinated dibenzo-*p*-dioxins (PCDDs; dioxins)
  - 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD)
    - Present in 2,4,5-T (herbicide)
    - Treatment: symptom-based

- Symptoms: dermatologic problems (chloracne dryness, rash, hyperpigmentation), liver dysfunction,
  - endocrine & neurologic dysfunction

QUICK REVIEW: OCCUPATIONAL TOX

- What part of the electron transport chain does rotenone inhibit?

  I

- What is glyphosate used for?

- What is the role of pralidoxime in treating organophosphate toxicity?

  reverses AChE inhibition

- How does hyperbaric O_2 treat CO poisoning?

  outcompete and replace so body can clear

QUICK REVIEW OCCUPATIONAL TOX ANSWERS

- What part of the electron transport chain does rotenone inhibit?
  - Complex I (I_Q binding site)
- What is glyphosate used for?
  - herbicide (chlorophenoxy-type; Roundup)
- What is the role of pralidoxime in treating organophosphate toxicity?
  - it acts as a cholinesterase "reactivator" by unconjugating (by cleaving the phospho-ester bond) the organophosphate from the enzyme
- How does hyperbaric O_2 treat CO poisoning?
  - high pO_2 replaces CO-Hg with O_2-Hg faster than normoxic or atmospheric O_2

LECTURE OUTLINE

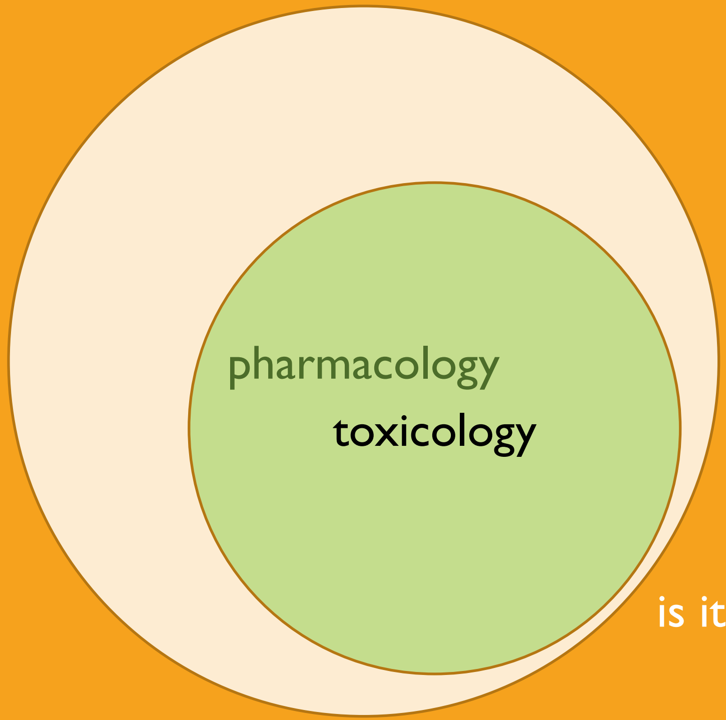
Drug lists (slides 4-5: on your own)

Drug toxicity: managing the poisoned patient (Katzung 58, Goldfrank's)

Individual common toxins (on your own)

Occupational Tox: Common air and envt pollutants (Katzung 56)

Heavy metals (Katzung 57)



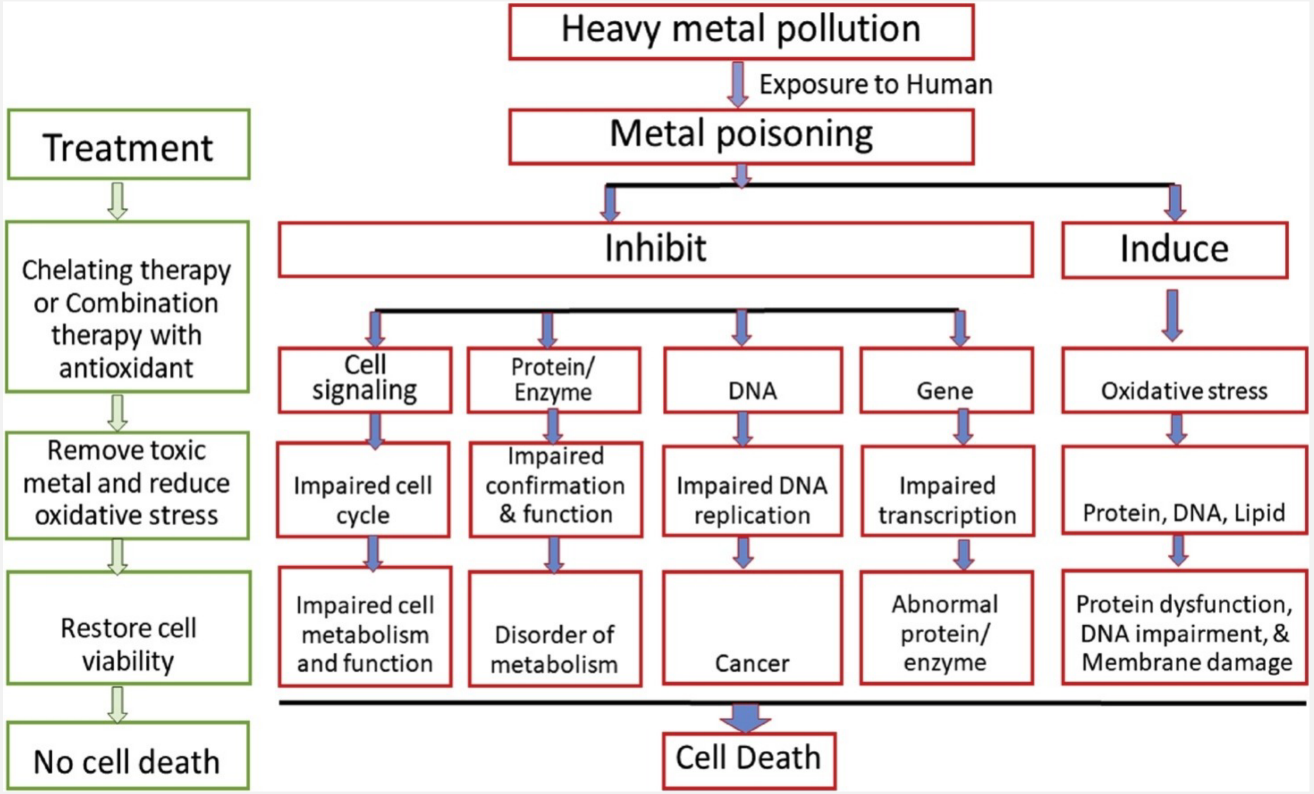
is it a little cramped in here?

METALS TOX TREATMENT MOA: METAL CHELATION

KNOW

Chelators

- Dimercaprol (BAL)
- Succimer (DMSA)
- Edetate calcium disodium (EDTA)
- Unithiol (DMPS)
- Penicillamine
- Deferoxamine
- Deferasirox
- Prussian blue



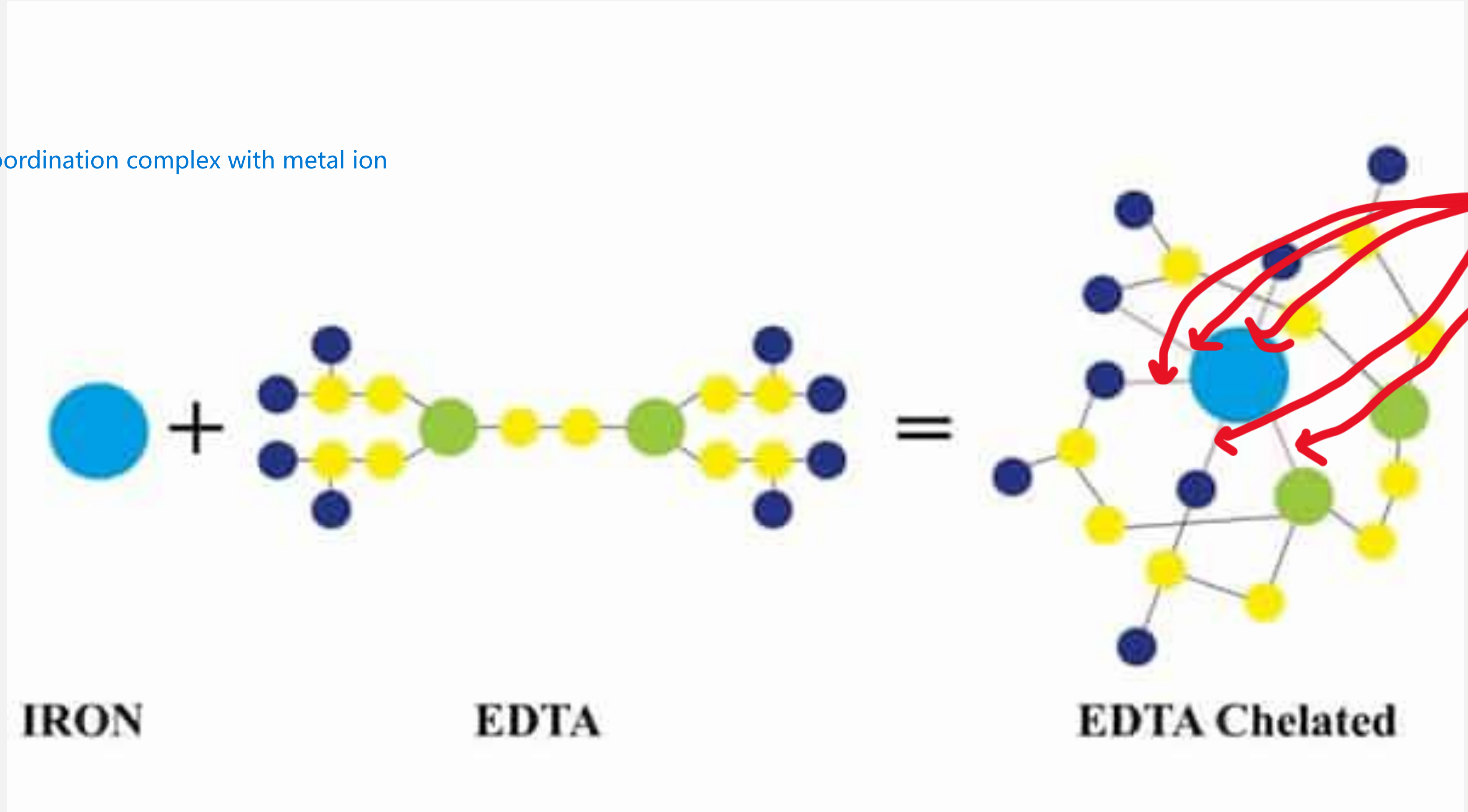
easy peasy! (?)

Common heavy metals

- **lead**
- **arsenic**
- **mercury**
- cadmium
- beryllium
- etc.

CHELATION: A REFRESHER

chelating agent makes coordination complex with metal ion



polydentate ligand: forms more than one bond to the same ion

1. review coordination complex chemistry in your favorite textbook.
2. Or, remember that a chelator forms a coordination complex with a metal ion

TOXICITY PROFILES OF DIFFERENT METALS

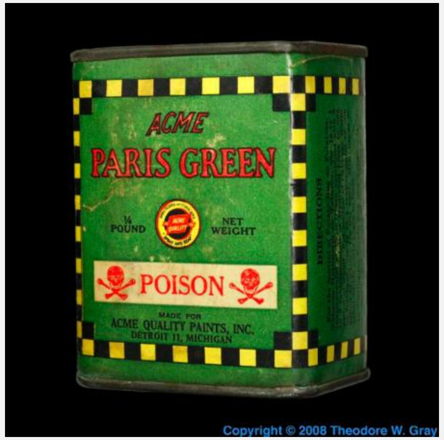
TABLE 57–1 (Katzung) Important characteristics of the toxicology of lead, arsenic, mercury, and iron.

Metal ion	Form Entering Body	Route of Absorption	**Target Organs for Toxicity**	**Treatment[a]**
Lead	Inorganic lead oxides and salts	Gastrointestinal, respiratory, skin (minor)	Hematopoietic system, CNS, kidneys	Dimercaprol, EDTA, succimer, unithiol
	Tetraethyl lead	Skin (major), gastrointestinal	CNS	Seizure control
Arsenic	Inorganic arsenic salts	All mucous surfaces	Capillaries, gastrointestinal tract, hematopoietic system	Dimercaprol, unithiol, succimer, penicillamine
	Arsine gas	Inhalation	Erythrocytes	Supportive
Mercury	Elemental	Inhalation	CNS, kidneys	Succimer, unithiol
	Inorganic salts	Gastrointestinal	Kidneys, gastrointestinal tract	Succimer, unithiol, penicillamine, dimercaprol
	Organic mercurials	Gastrointestinal	CNS	Supportive
Iron	Ferrous sulfate	Gastrointestinal	Gastrointestinal, CNS, blood	Deferoxamine, deferasirox, deferiprone

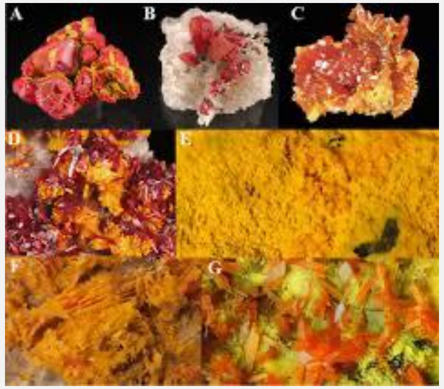
COMMON ROUTES OF EXPOSURE

- Industrial: machining/manufacturing/agriculture/etc.
- Envt contamination of food: bioaccumulation in e.g., fish, products irrigated in contaminated water
- Spices, cocoa: FDA-regulated (as food), but little routine testing
  - https://www.consumerreports.org/health/food-safety/your-herbs-and-spices-might-contain-arsenic-cadmium-and-lead-a6246621494/
- Pigments/paints, cosmetics, jewelry
- Natural supplements: unregulated by the FDA, so testing not routine
  - https://www.sciencedirect.com/science/article/pii/S2214750021000883

anything milled into a powder can be contaminated



Cu_3As_2

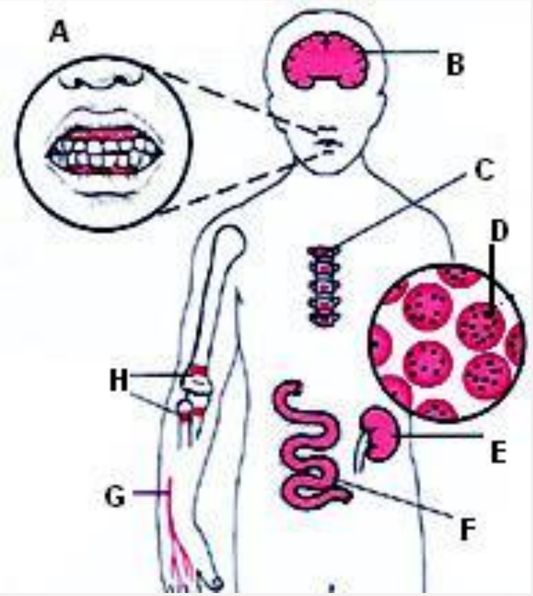


As_2S_3

it's too bad arsenic makes such great pigments

LEAD EXPOSURE

- *No* safe threshold established
- Absorbed through *respiratory* and GI tracts (inorganic lead)
- Bound to *erythrocytes*, distributed to *soft tissues* and *bone*
- Crosses the *placenta* (**fetus/young children are most sensitive**)
- Eliminated in urine
- **Causes encephalopathy** (behavior/neurocognitive effects),
  - **hypertension, anemia, basophilic stippling** (RBC),
  - **nephropathy, reproductive dysfunction, GI irritation**
  - **'lead colic', 'gingival lead lines', wrist drop**

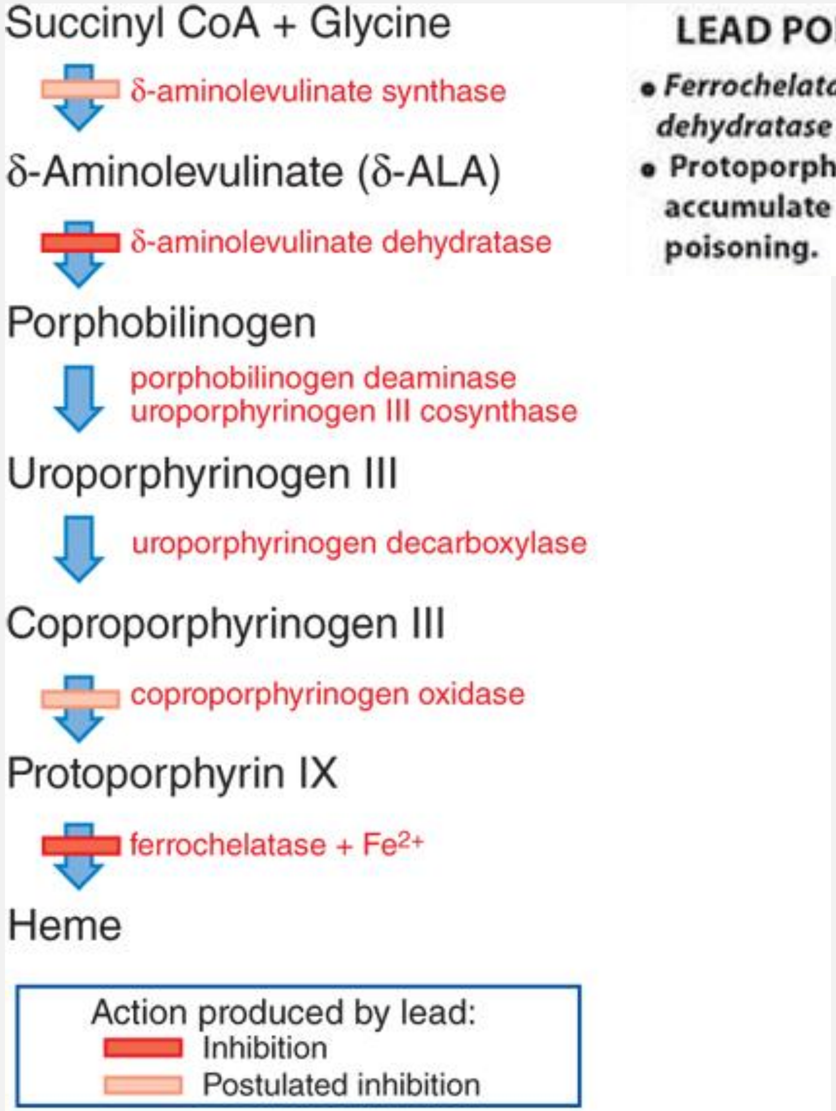


- **Treatment:** *Immediate termination of exposure*
  - **Supportive care: corticosteroids and mannitol for *cerebral edema*; anticonvulsant for seizures, GI decontamination**
  - **Chelation: edetate calcium disodium (IV; $CaNa_2EDTA$)**

study note: note that chelation is not always indicated; depends on degree of exposure

ANEMIA: MORPHOLOGICAL CLASSIFICATION

- Macrocytic-normochromic anemias
  - Pernicious (Vit B12 deficiency)
  - Folate-deficiency
- Microcytic-hypochromic anemias
  - Iron-deficiency
  - Sideroblastic
  - Thalassemia
  - Chronic disease
  - **Lead poisoning**
- Normocytic-normochromic anemias
  - Aplastic
  - Posthemorrhagic
  - Hemolytic
  - Sickle cell



LEAD POISONING

- ***Ferrochelatase*** **and** ***ALA dehydratase*** **are inhibited by lead.**
- **Protoporphyrin and ALA accumulate in the urine in lead poisoning.**

FYI anemia Hb:
M: < 13 g/dL
F: < 12.5 g/dL

HEALTH EFFECTS OF LEAD

BLL (μg/dL)	HEALTH EFFECT
Children	
<10	Inhibition of neural development Increased blood pressure Hearing loss Inhibition of hemoglobin synthesis enzymes Short stature Delayed sexual maturation (girls)
>10	Immunological effects (increased IgE)
>15	Increased erythrocyte protoporphyrin Decreased vitamin D
>30	Depressed nerve conduction velocity
>40	Anemia
>45	**Chelation therapy warranted**
>60	Colic
>70	Encephalopathy

BLL (μg/dL)	HEALTH EFFECT
Adults	
<10	Increased blood pressure Inhibition of hemoglobin synthesis enzymes Decreased glomerular filtration
>20	Increased erythrocyte protoporphyrin
>30	Hearing loss Enzymuria/proteinuria
>40	Peripheral neuropathy Reduced fertility Altered thyroid hormone levels Neurobehavioral effects
>50	Anemia
>60	Colic **Chelation therapy warranted**
>100	Encephalopathy

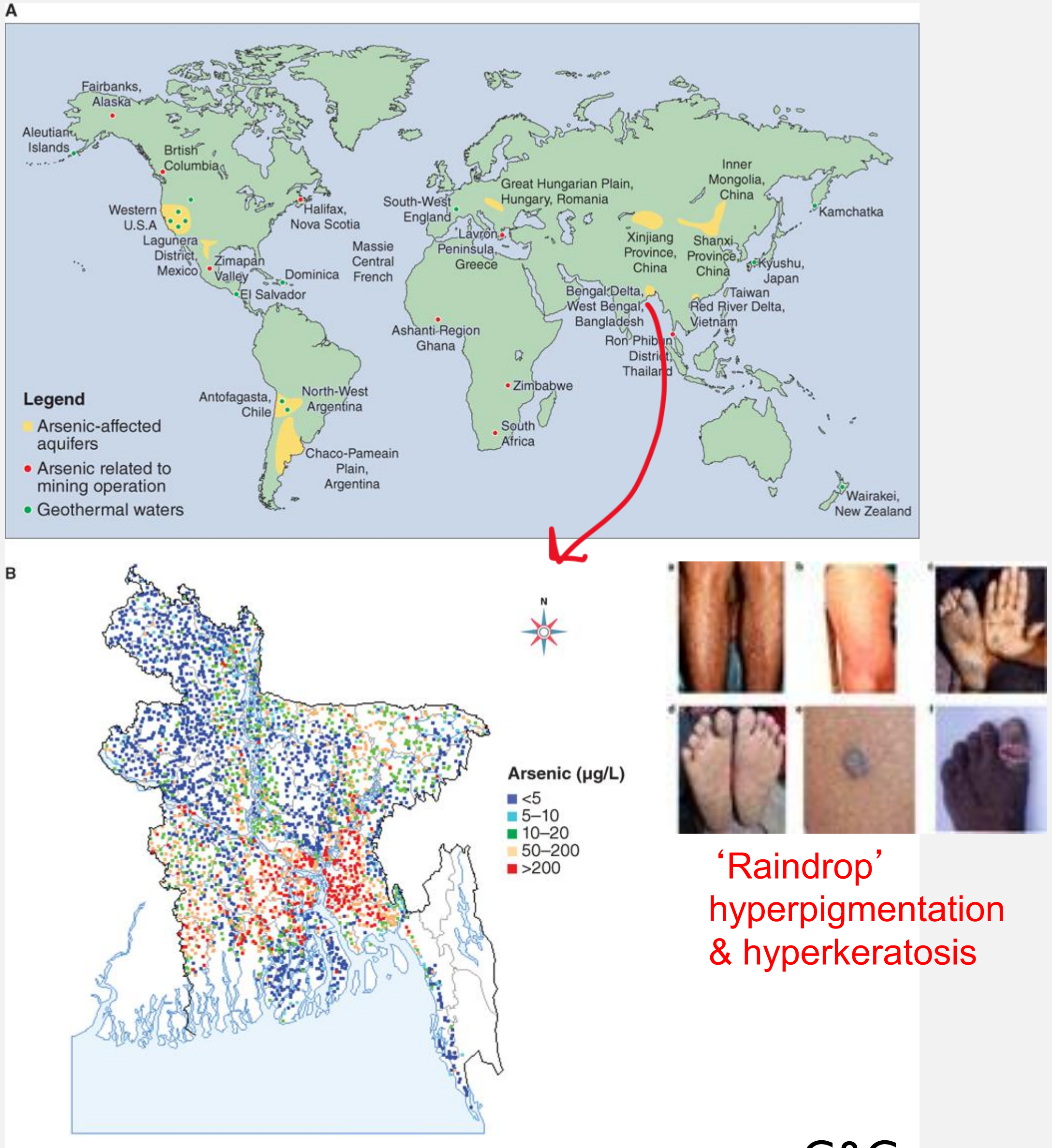
FYI specific thresholds; know general symptoms and BLL (blood lead level) when chelation therapy indicated

G&G

ARSENIC: A NATURALLY OCCURRING ELEMENT

- naturally-occurring metalloid
  - widely distributed, but
  - concentrated in gold-containing ores
  - released by gold mining operations

- routes of exposure
  - most common: drinking water, food
    - rice and rice products of concern in baby foods, e.g., rice cereals and brown rice syrup
  - also happens: As-containing pesticides, metallurgy, semiconductor mfr, pressure/”weather”-treated wood, As-containing pigments

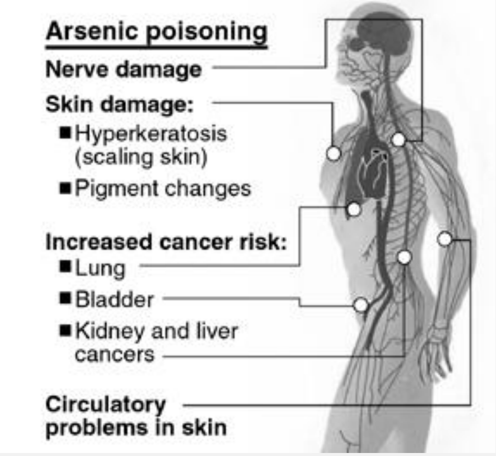


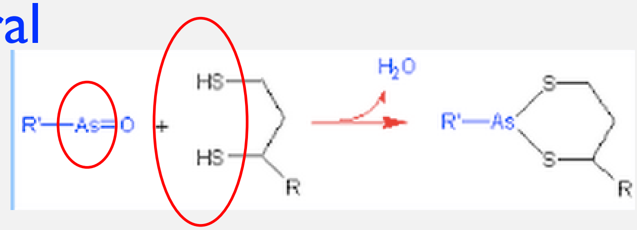
‘Raindrop’ hyperpigmentation & hyperkeratosis

G&G

SYMPTOMS OF ARSENIC EXPOSURE

- **Arsenic** (3+ or 5+)
- Absorbed through the respiratory and GI tracts
- Metabolized to *methylated* metabolites in liver, excreted in urine
- Bound to sulfhydryl (= thiol, -SH) groups present in keratinized tissues hair and nails
- **Inhibits sulfhydryl-containing *enzymes* (e.g.,G6PD)**
- Arsenate ($AsO4^{3-}$) mimics phosphate, thus interrupts glycolysis
- Toxicity: **As^{3+} > As^{5+} ; Inorganic As > organic As**
  - carcinogen, skin damage, neuropathy
- Symptoms: GI symptoms, hypotension, cardiopulmonary toxicity, encephalopathy, delirium, coma, peripheral neuropathy, Aldrich-Mees lines in the nails
- Some trimethylated organoarsenic are *not* toxic (arsenobetaine, arsenosugars are found in seafood)



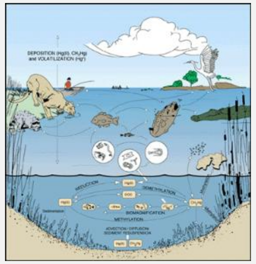


ARSENIC EXPOSURE: TREATMENT

- **Arsenic**
- **Treatment:**
  - Acute arsenic poisoning (GI, hypotension, cardiac dysfunction)
    - Gut decontamination, supportive care, chelation therapy
      - preferred chelators: **dimercaprol** (IM), **succimer,** or unithiol (IV) (analog of dimercaprol, not approved in US)
      - clinical notes on dimercaprol: low therapeutic index, formulated with peanut oil
  - Chronic arsenic poisoning (*skin changes*, anemia, sensorimotor
  - peripheral neuropathy, hypertension, cancer)
    - Supportive care, chelation therapy (succimer)
  - Arsine gas poisoning (*hemolytic effects*, malaise, headache,
  - GI symptoms, jaundice, hemoglobinuria, renal failure)
    - Supportive care, hemodialysis (for renal failure)

MERCURY EXPOSURE

- **Mercury**
- Found in electrical equipment, fluorescent lamp bulbs, thermometers, dental amalgam, skin "brighteners", novelty/costume jewelry, red pigments (cinnabar)
- Organic (methylmercury most toxic), inorganic, element (no GI abs., vapor can be inhaled)
- *Biomagnification* of *methylmercury* in *fish*
- Absorbed through *inhalation*, skin contact
- Inhibit *sulfhydryl-containing enzymes* (e.g., G6PD and glutathione reductase) and alter *cell membranes*
- Eliminated in urine and feces
- **Symptoms**:
  - vision impairment,
  - disturbances in sensations
  - (numbness in the foot/hand or mouth),
  - lack of coordination of movements,
  - impairment of speech, hearing, walking;
  - muscle weakness, skin rashes,
  - memory loss, mental disturbance
  - developmental abnormalities

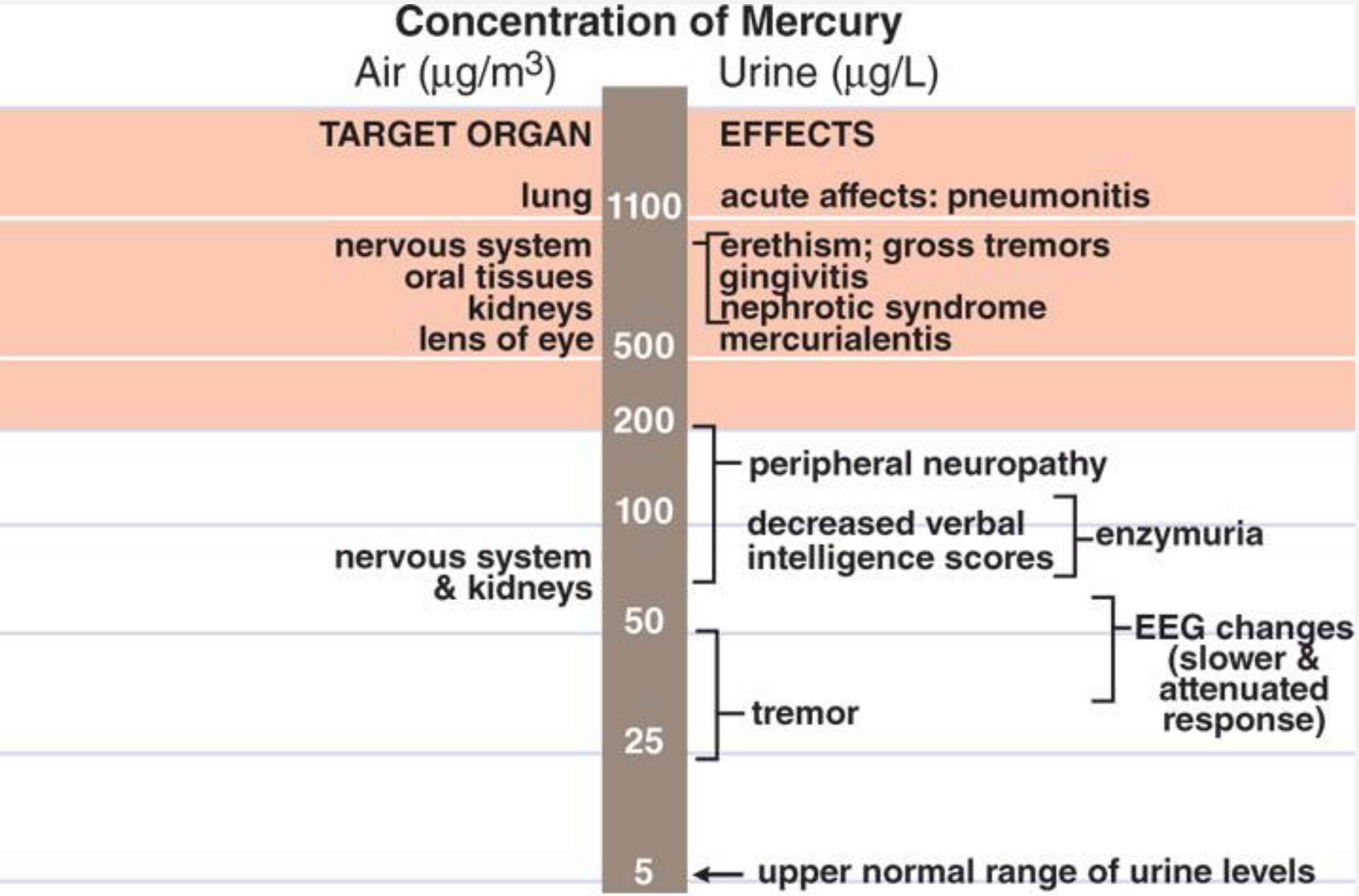



'Minamata disease'
(Japan, 1932-1968)

FYI: industrial dumping of mercury waste into Minamata coastal waters by the Chisso Corporation Litigation not resolved until 2004

TREATING MERCURY EXPOSURE

- **Treatment**
  - **Acute** (chemical pneumonitis, noncardiogenic pulmonary edema, neurologic sequelae, life-threatening hemorrhagic gastroenteritis followed by renal failure)
    - Supportive care, chelation therapy (unithiol, **dimercaprol, succimer**) and hemodialysis
  - **Chronic** (triad of tremor, neuropsychiatric disturbance-erethism, gingivostomatitis)
    - Unithiol and succimer: ↑ mercury excretion
    - ***Avoid*** **dimercaprol** in chronic Hg exposure (*redistributes* Hg to CNS)



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GENERAL PHARMACOLOGY OF CHELATORS

- **MOA: forms a coordinate complex with metal ions, preventing biological interaction**
- ***Accelerates* the elimination of metal ions from the body**
- **SE:** May also enhance the elimination of *essential cations* (Zn, Cu)
- Usually contain *electronegative* groups (-SH,-OH,-NH) that form stable *coordinate covalent* bonds with cation metal atom
- Most effective when administered soon after an acute metal exposure

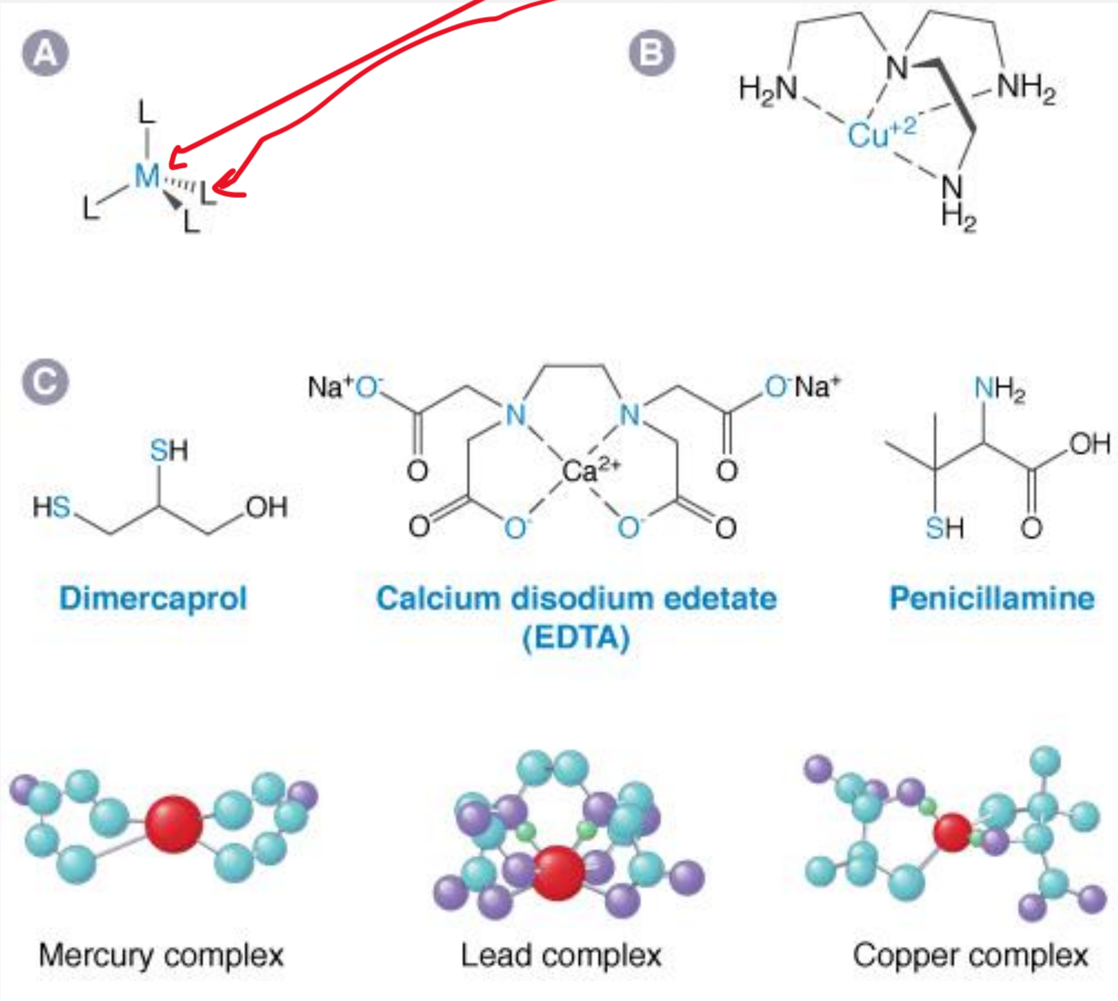
important clinical note! EDTA is a strong Ca^{++} chelator: USE Ca^{++}-EDTA, NOT $2Na^{+}$-EDTA, to avoid hypocalcemia

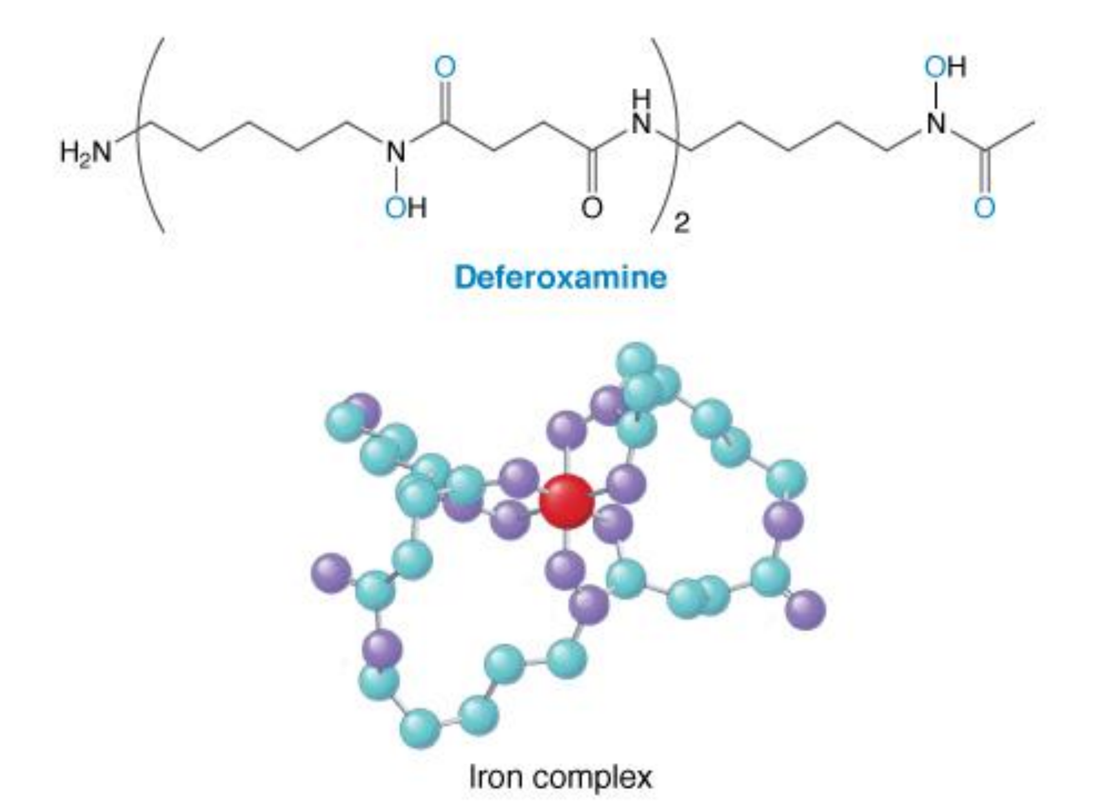
- Dimercaprol (BAL)
- Succimer (DMSA)
- Unithiol (DMPS)
- Edetate **calcium** disodium (EDTA)
- Penicillamine
- Deferoxamine
- Deferasirox
- Prussian blue

just because it's natural doesn't mean it's safe

CHELATORS "SURROUND" METAL IONS

complexation nomenclature: "metal" or "coordination center" plus "ligands"



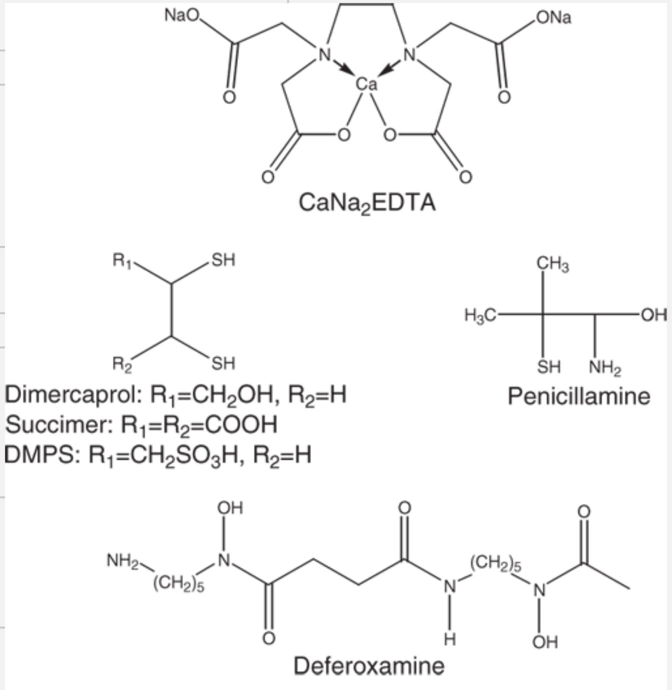


ligands are usually thiol or nitrogen groups on chelators

COMMON CHELATORS: INDICATIONS, SE

Drugs	Therapeutic Uses	Clinical Pharmacology and Tips
Heavy Metal Chelators		
$CaNa_2EDTA$ (calcium disodium ethylenediaminetetraacetic acid)	Acute lead poisoning	•IV or IM administration •Not effective for chronic lead poisoning •Nephrotoxic •IV administration can increase intracranial pressure in patients with lead encephalopathy and cerebral edema •Zinc supplementation may be beneficial
Dimercaprol	•Acute arsenic, gold, and mercury poisoning •Acute lead poisoning (in combination with $CaNa_2EDTA$)	•Administered IM •Not effective for chronic intoxication •Increases toxicity of iron, cadmium, and selenium •Toxicity profile is worse than for succimer
Succimer	•Treatment of children with blood lead > 45 µg/dL •Off label for adults with lead poisoning and for arsenic and mercury poisoning	•Orally bioavailable •Improved toxicity profile over dimercaprol •Reduced mobilization of lead to brain •May cause allergic reactions
Copper Chelators		
Penicillamine	•Treatment of copper intoxication due to Wilson disease •Chelates heavy metals, but more toxic, less potent, and less selective than other options	•Orally bioavailable •Allergenic •Nephrotoxic •Causes hematological toxicities •Causes a variety of other side effects
Trientine	Treatment of Wilson disease in those intolerant of penicillamine	•Orally bioavailable •Less potent than penicillamine
Iron Chelators		
Deferoxamine	•Treatment of acute iron intoxication •Treatment of chronic iron overload due to transfusion	•IV, IM, or SC administration required •SC administration preferred for chronic iron overload •IV use for cardiovascular collapse or shock •IM administration for other acute iron intoxication cases
Deferasirox	•Treatment of chronic iron overload due to transfusion •Treatment of nontransfusion-dependent iron overload	•Orally bioavailable •Renal failure, hepatic failure, and GI hemorrhage are concerns •Not recommended over deferoxamine
Deferiprone	Treatment of chronic iron overload due to transfusion	•Orally bioavailable •Causes agranulocytosis and neutropenia •Not recommended over deferoxamine

take-home: note in what contexts (metal, timeframe) agents are effective or ineffective; speculate about why



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CHELATORS, CONT'D

- **Dimercaprol** (British Anti-Lewisite or **BAL**)

- Unstable (oxidized rapidly)
- Water-insoluble: dispensed in peanut oil; **IM**
- Used to treat acute intoxication by **arsenic**, **lead** (+ EDTA), **mercury**
- Increase the *excretion* of arsenic and lead
- Side effects: hypertension, tachycardia, nausea, vomiting,
- lacrimation, salivation, fever, & pain at injection site,
- ↑prothrombin time, thrombocytopenia
- *Redistribute* arsenic and mercury to **CNS** (*avoid* in chronic poisoning)
- **Water-soluble analogs: succimer, unithiol**

- **Succimer** (Dimercaptosuccinic acid, **DMSA**)

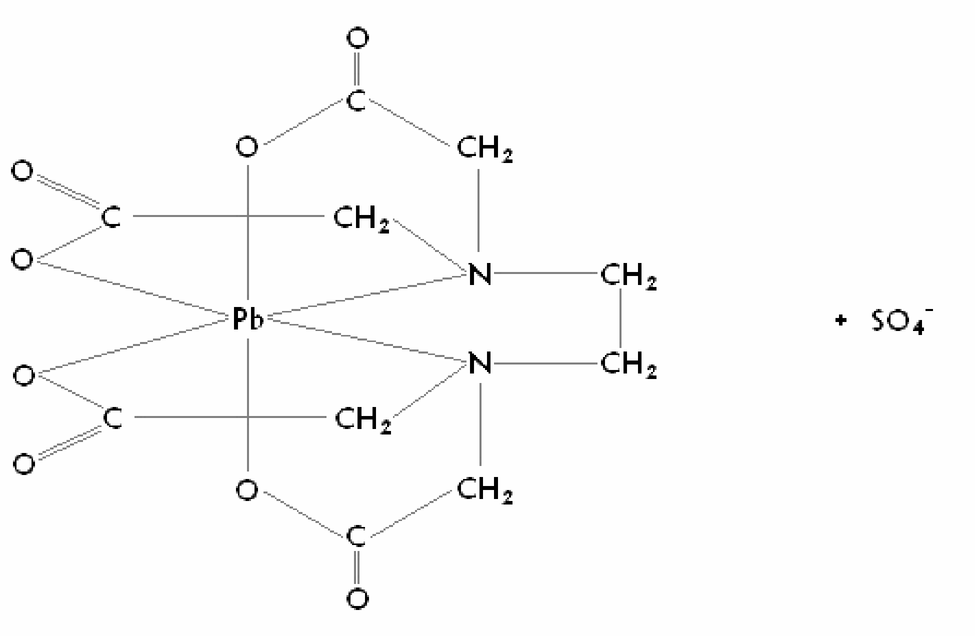
- water-soluble
- Oral use in USA only
- ↑ lead excretion; ↓ kidney mercury
- Approved for treating lead poisoning in children
- Also used to treat arsenic and mercury poisoning
- Side effects: GI irritation, rash, ↑liver enzymes

CHELATORS, CONT'D

- **Unithiol** (Dimercaptopropanesulfonic acid, **DMPS**)

- Oral or IV (infused slowly to avoid vasodilation & hypotension)
- ↑ excretion of mercury, lead and arsenic
- Side effects: dermatologic rxs (urticaria) are most common, may
- cause major allergic rxs (e.g., SJS)
- **_Edetate calcium disodium_** (Ethylenediaminetetraacetic acid, **EDTA**)

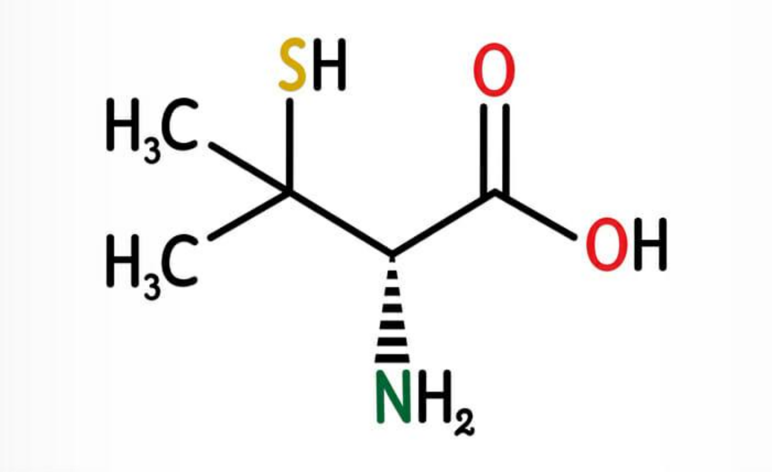
- Used as the calcium disodium salt (to *avoid depletion* of Ca)
- Highly polar structure (poor oral absorption & membrane penetration)
- *Mobilize* lead from soft tissues (↑ lead excretion, ↓ blood lead)
- Excreted in urine
- Used mainly to chelate lead
- Caution in *anuric* patients will cause accumulation of lead and EDTA in urine
- Side effects: nephrotoxicity (maintain hydration), zinc depletion
- **DTPA** (analog of EDTA) is used for removal of certain
  - *radioisotopes* (e.g., uranium, plutonium)



DTPA: diethylenetriaminpentaacetic acid

CHELATORS, CONT'D

- **Penicillamine** aka cupermine
- Derivative of penicillin
- D-penicillamine is preferred (less toxic than L-isoform)
- Well absorbed by gut; resistant to metabolic degradation
- Mainly used for prevention or treatment of copper poisoning
- (e.g., Wilson's disease)
- ↑ urinary excretion of lead and mercury (replaced by succimer)
- Also used to treat RA
- Side effects: hypersensitivity rxs (rash, purities), nephrotoxicity
- with proteinuria, pyridoxine deficiency



Penicillamine

a.k.a., cupramine

- **Deferoxamine/deferasirox**
- Used for iron poisoning or iron overload by blood transfusion
- Deferoxamine: IV or IM
  - Chelated cplx excreted in urine (orange-red urine)
- Deferasirox: orally active
  - Chelated cplx excreted in bile
- Long-term use may cause neurotoxicity and ↑ infections

CHELATORS, CONT'D

- **Prussian blue** **(ferric hexacyanoferrate)**

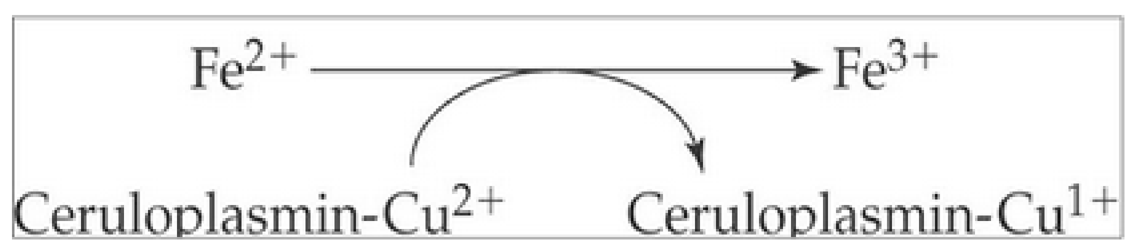
- A dark-blue pigment; *insoluble* compound is used as a chelator
- High affinity (mainly through ion exchange) for certain
- univalent cations (esp cesium, thallium)
- Chelated complex are nonabsorbable; ↑excretion in feces
- Approved for treatment of contamination with ^{137}Cs and for
- intoxication with thallium salts
- Side effects: constipation

what makes "blueprints" blue?
yes, you guessed right!

rainy day reading hole: google "cyanotype chemistry"

COPPER: DERANGEMENT IN SPECIFIC DISEASES

- **Wilson’s disease**
  - **Autosomal recessive loss of function mutations in ATPase (ATP7B)**
  - **Result in defective copper excretion into bile and hence copper toxicity.**
  - **Typical presentation is <40-yo with liver disease or neuropsychiatric disorder. Kayser–Fleischer corneal rings may be seen.**
  - **Low serum ceruloplasmin, high urinary copper excretion, and elevated liver copper content**
  - **Treatment is with copper chelating agents and zinc**



- **Menkes’ disease**
  - X-linked loss of function mutations in ATP7A transporter (trans golgi apparatus)
  - result in defective copper transport across intestine, placenta and brain and hence cellular copper deficiency.
  - Clinical presentation is in infancy with facial dimorphism, connective tissue disorder, hypopigmentation, abnormal hair, seizures and failure to thrive, usually followed by death by age 3 years.
  - Treatment is with intravenous **copper histidine.**

QUICK REVIEW: METALS

- T/F: heavy metal exposure requires treatment with metal chelators

  F

- How does lead exposure cause anemia?

- What side group does As bind to in enzyme targets?

- Which is water-soluble, dimercaprol or succimer?

- How does a chelator interact with a metal ion?

  surrounds it

3:00

QUICK REVIEW # 2 ANSWERS

- T/F: heavy metal exposure requires treatment with metal chelators
  - F; for some metals and exposures, chelation therapy not indicated

- How does lead exposure cause anemia?
  - inhibiting delta-ALA and ferrochelatase

- What side group does As bind to in enzyme targets?
  - thiol (sulfhydryl)

- Which is water-soluble, dimercaprol or succimer?
  - succimer

- How does a chelator interact with a metal ion?
  - forms a coordination complex

THANKS!!!

- smookerj@touro.edu

- please reach out by email with questions or to schedule individual/group office time
  - response will be faster during normal business hours

PRACTICE QUESTIONS

1 A 56-year-old man who worked in the coal mines for many years was recently diagnosed with pulmonary fibrosis. This is an example of:

A Acute toxicity.
B Chronic toxicity.
C Epigenetic alterations.
D Genetic alterations.
E Immunotoxicity.

2 Bisphenol A (BPA) exposure can affect both DNA methylation and histone modification. These changes are known as:

A Genetic alterations.
B Developmental alterations.
C Epigenetic alterations.
D Cell-specific alterations.
E Nonheritable alterations.

3 The leading cause of death by poisoning is due to exposure to:

A Benzene.
B Chloroform.
C Paraquat.
D Bisphenol A (BPA).
E Carbon monoxide.

4 Which of the following correctly pairs a toxin with an associated abnormality or disease?

A Tobacco smoke—cancer.
B Formaldehyde—teratogenesis.
C Chloroform—hepatic cirrhosis.
D Cadmium—nephrotoxicity.
E All of the above.

5 The strongest evidence for an association between environmental toxicants and a neurodegenerative disorder exists for which disorder?

A Multiple sclerosis.
B Alzheimer disease.
C Parkinson disease.
D Amyotrophic lateral sclerosis.
E Myasthenia gravis.

Want answers and more explanation? Source: Brody's Human Pharmacology

PRACTICE QUESTIONS

1. Calcium disodium EDTA is employed in the treatment of which metal poisoning?
   (A) lead
   (B) cadmium
   (C) mercury
   (D) arsenic
   (E) beryllium

2. Which herbicide can cause renal failure after accidental or intentional ingestion?
   (A) 2,4-dichlorophenoxyacetic acid (2,4-D)
   (B) glyphosate
   (C) paraquat
   (D) diquat
   (E) diazinon

3. What is the most characteristic feature of chronic mercury poisoning?
   (A) hypothermia
   (B) heart failure
   (C) biliary obstruction and jaundice
   (D) anger, depression, irrational behavior
   (E) rhabdomyolysis

4. Which agent is used to reduce the gastrointestinal absorption of ingested poisons?
   (A) succimer
   (B) activated charcoal
   (C) pralidoxime
   (D) dimercaprol
   (E) unithiol

5. An increased risk of which disease has been linked to exposure to 2,4-dichlorophenoxyacetic acid (2,4-D)?
   (A) hypertension
   (B) hyponatremia
   (C) pulmonary fibrosis
   (D) dementia
   (E) non-Hodgkin lymphoma

Want answers and more explanation? Source: Brenner and Stevens' Pharmacology

CASE STUDY

- A 6-year-old girl is brought to the emergency department by her parents*. She is comatose, tachypneic (25 breaths per minute), and tachycardic (150 bpm), but she appears flushed, and fingertip pulse oximetry is normal (97%) breathing room air. Questioning of her parents reveals that they are homeless and have been living in their car (a small van). The nights have been cold, and they have used a small charcoal burner to keep warm inside the vehicle.
- What is the most likely diagnosis?
- What treatment should be instituted immediately?
- If her mother is pregnant, what additional measures should be taken?

Katzung

- **CASE STUDY ANSWER**
- The child presents with classic signs (and history) of **carbon monoxide (CO) exposure**. Pulse oximetry is unreliable in CO poisoning, although newer instruments may distinguish between carboxyhemoglobin (CO-Hgb) and oxyhemoglobin.
- Institute the ABCDs of poisoning (see Chapter 58). Immediate high-flow oxygen is mandatory and should be administered via a tight-fitting face mask or endotracheal catheter. A blood sample for blood gases and carboxyhemoglobin content should be obtained. If the CO-Hgb is greater than 50%, hyperbaric oxygen treatment (if available) may be considered. The electrocardiogram should be continuously monitored for arrhythmias. Anticonvulsant drugs may be required if seizures occur. Neurologic damage due to CO exposure may be subtle and long-lasting; the child should be followed for years if necessary.
- The fetus is particularly susceptible to hypoxia, and if the mother is pregnant, her blood gases and CO-Hgb should be measured. If the latter is high, hyperbaric oxygen therapy should be considered.

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OTHER CASE STUDY RESOURCES: FYI

- HIGHLY recommend: https://www.atsdr.cdc.gov/csem/csem.html